



Cyber Security of Power Grids

Dr. Alex Stefanov

MSc EE & MSc SET, Course EE4410

Faculty of Electrical Engineering, Mathematics and Computer Science, **TU Delft**

Let's Introduce Ourselves ...

- **Assistant Professor**, Intelligent Electrical Power Grids at EEMCS, **TU Delft**
 - Cyber security of power grids
 - Resilience of transmission & distribution systems to cyber attacks



Source: marketingtwentyone.co.uk

Cyber Security of Power Grids: Objectives

Course aimed to analyse and enhance cyber security of power grids

- Analyses typical Information Technology (IT) – Operational Technology (OT) system architectures used for power grid monitoring and control
- Explains main cyber security concepts

Course provides essential knowledge to

- Analyse cyber security threats and vulnerabilities of power grids
- Examine kill chain stages of cyber attacks on power grids using a real-world study case
- Study state-of-the-art security controls and cyber security best practices to defend against cyber attacks



Cyber Security of Power Grids: Learning Objectives

At the end of the course, you will be able to

1. Analyse cyber security threats, threat vectors, and vulnerabilities of IT–OT systems for power grids
2. Examine kill chain stages of the cyber attack process on cyber-physical power systems
3. Analyse the impact of cyber attacks on power system operation
4. Recommend appropriate security controls to detect, prevent and mitigate cyber attacks, and cyber secure power grids



Learning Objectives

Disclaimer & Ethics

1. **All information provided in the course is for educational purposes only.** The lecturer(s) and teaching assistant(s) are not responsible for any misuse of the information.
2. The course does not promote and is strongly against illegal activities such as unethical hacking, cracking and piracy.
3. **Do not attempt to violate the law with anything contained in the course.** Do not abuse the material provided and be responsible.
4. By participating in the course, **you (student) must accept that you will only lawfully use the material and knowledge you got from this course.**
5. Any actions and / or activities related to the material contained within this course are solely your (student) responsibility. **The misuse of the information in this course can result in criminal charges brought against the persons in question.**
6. The lecturer(s) and teaching assistant(s) will not be held responsible in the event any criminal charges are brought against any individuals misusing the information from this course to break the law.



Introduction to Computer Networks

Dr. Alex Stefanov

MSc EE & MSc SET, Course EE4410

Faculty of Electrical Engineering, Mathematics and Computer Science, **TU Delft**

Lecture 1 on April 22, 2025

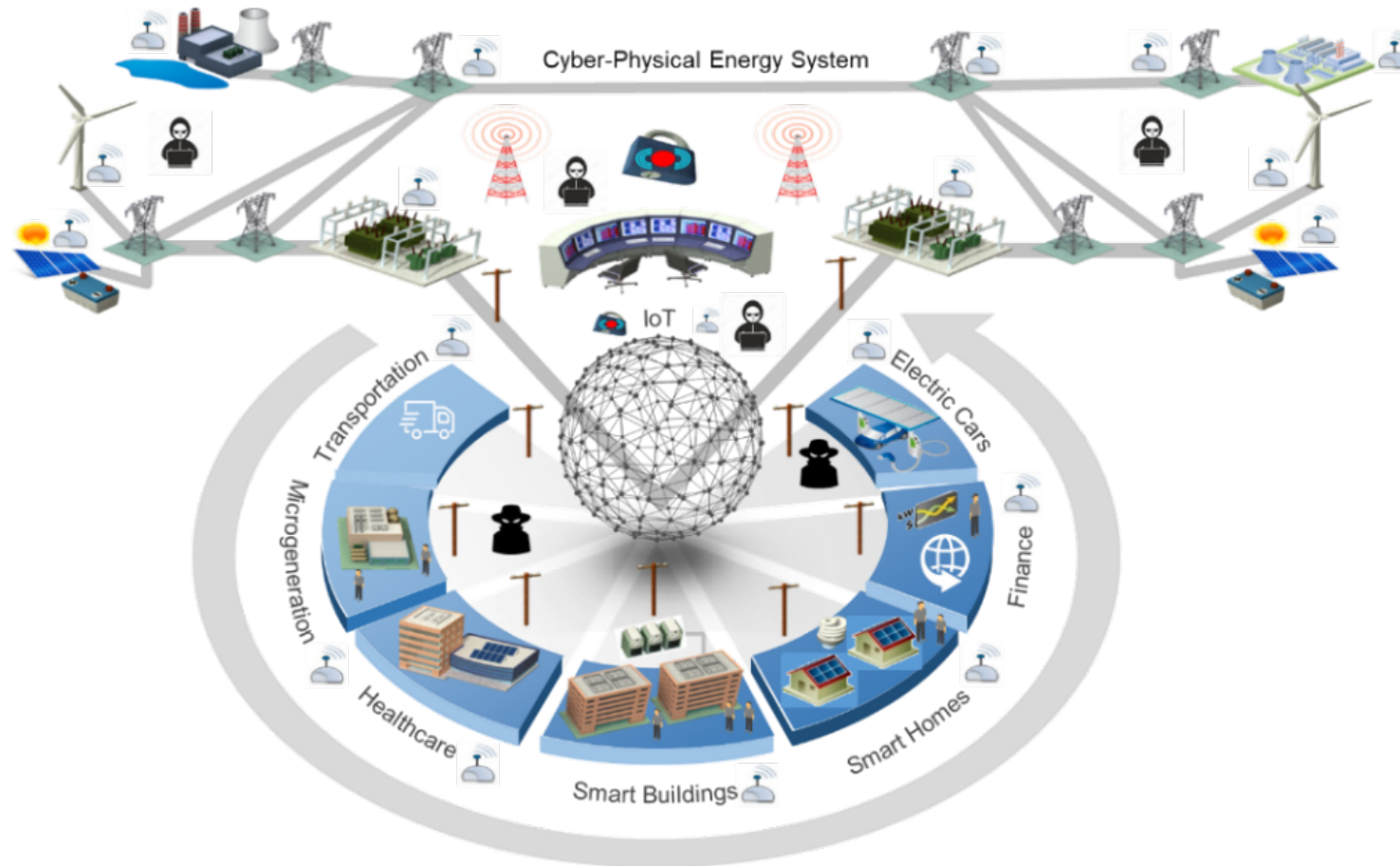
Power Grid at Heart of Energy Transition



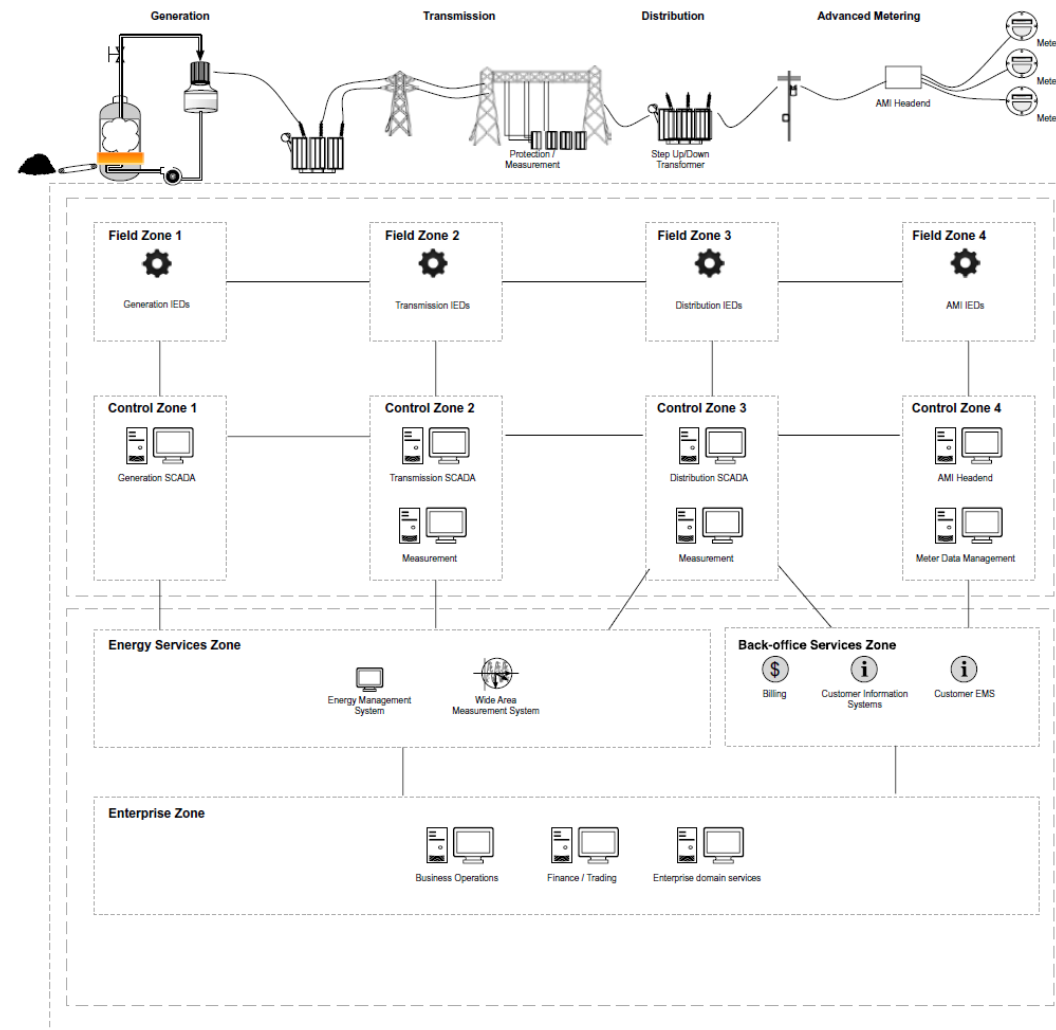




Smart Grids and Cyber Security

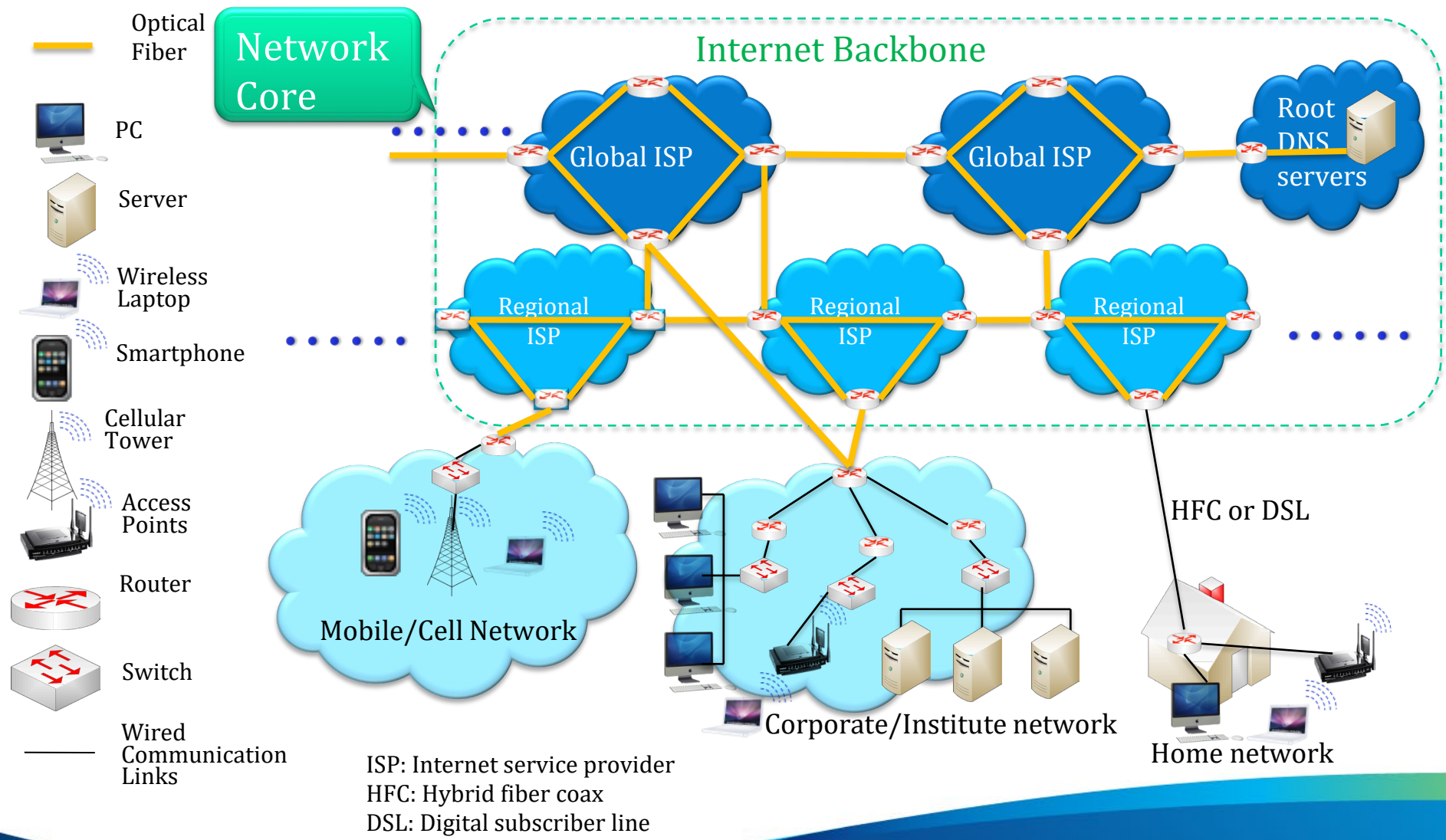


Supervisory Control and Data Acquisition (SCADA) Systems for Power Grids



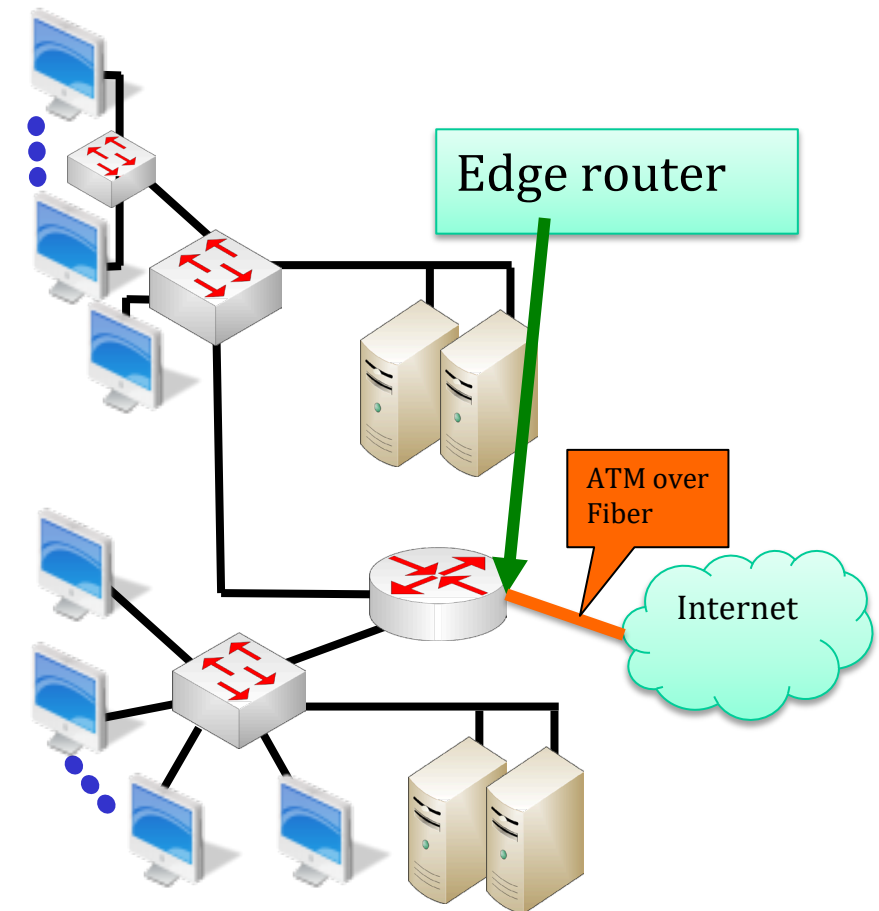
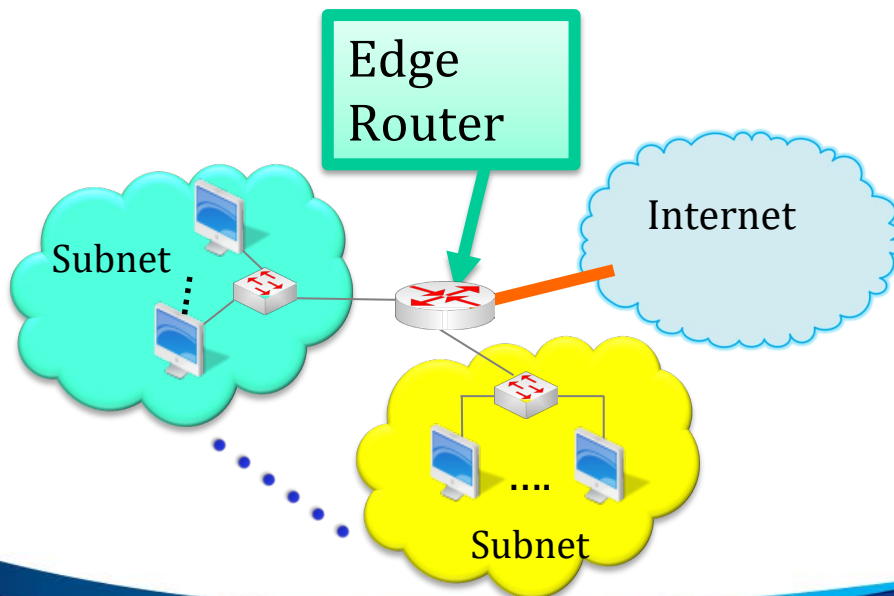
[Knapp, E. *et. al.* (2013). Applied cyber security and the smart grid: implementing security controls into the modern power infrastructure. Syngress]

Computer Networks: The Internet Architecture



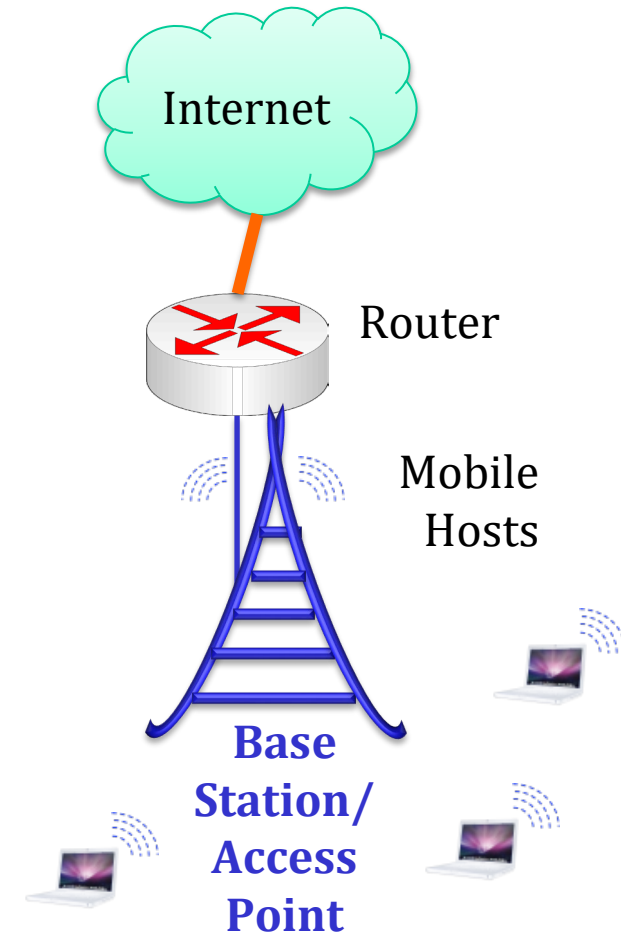
Computer Networks: Access Networks (Router and Subnet)

- Internet uses gateways (routers) to connect Local Area Networks (LAN)s / subnets to hierarchical network
 - Organization/home LAN or subnet connects hosts to edge router
 - Edge router connects LANs to Internet
 - ✓ Telco uses Asynchronous Transfer Mode (ATM) over fiber
 - Ethernet LAN
 - ✓ Hosts connect into Ethernet switch
 - ✓ 100Mbps, 1Gbps, 10Gbps, 100Gbps Ethernet



Computer Networks: Wireless Access Networks

- Shared wireless access network connects hosts to router
 - Via base station or access point
- Wireless LANs
 - 802.11a/b/g (WiFi): 11 or 54 Mbps
 - 802.11n > 100 Mbps
 - 802.11ac > 430 Mbps
- Wider-area wireless access
 - Provided by telco operator
 - 3G, 4G, 5G

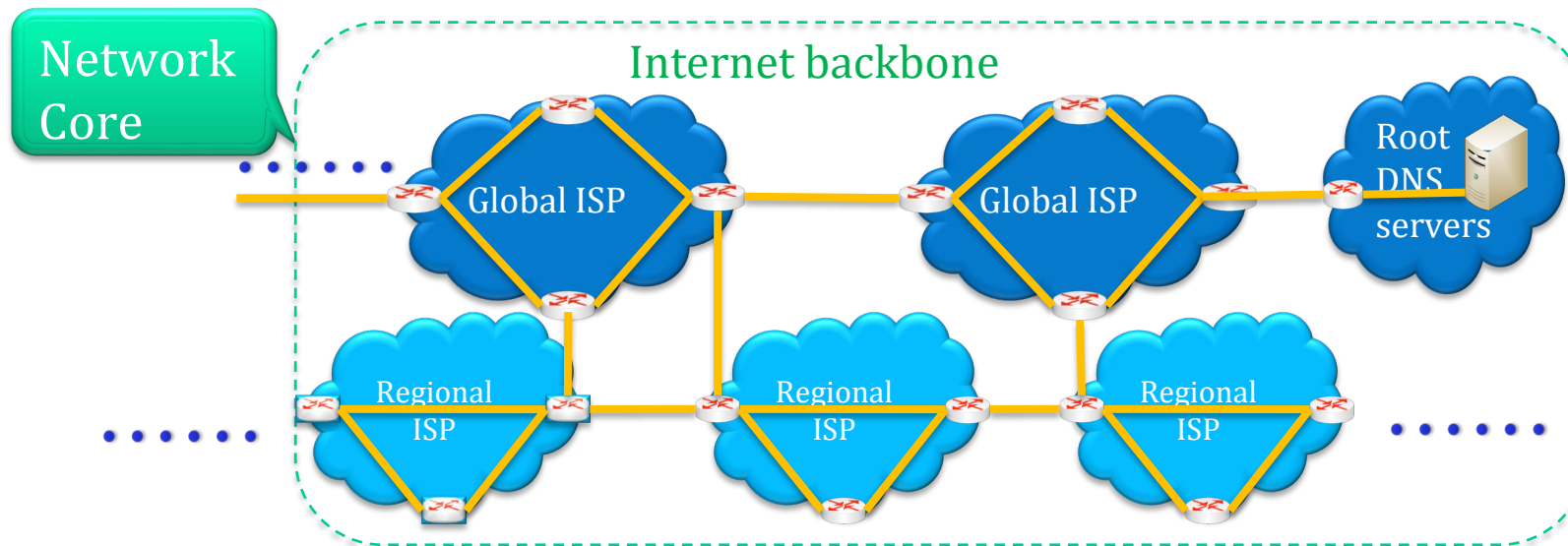


Computer Networks: Transmission (Physical) Media

- Physical link: medium between transmitter & receiver
- Transmission line (wire): signals propagate in a medium
 - Twisted copper pair: Ethernet 100BASE-T, 1000BASE-T, 10GBASE-T
 - Coax: 10BASE2
 - Fiber: 100BASEF, 1000BASEX, 10GBASE-R
- Free space: signals propagate in free space, e.g., radio waves
 - Wireless LAN: 802.11
 - 3G / 4G / 5G wireless
 - Long-Term Evolution (LTE), Worldwide Interoperability for Microwave Access (WiMAX)
 - Satellite

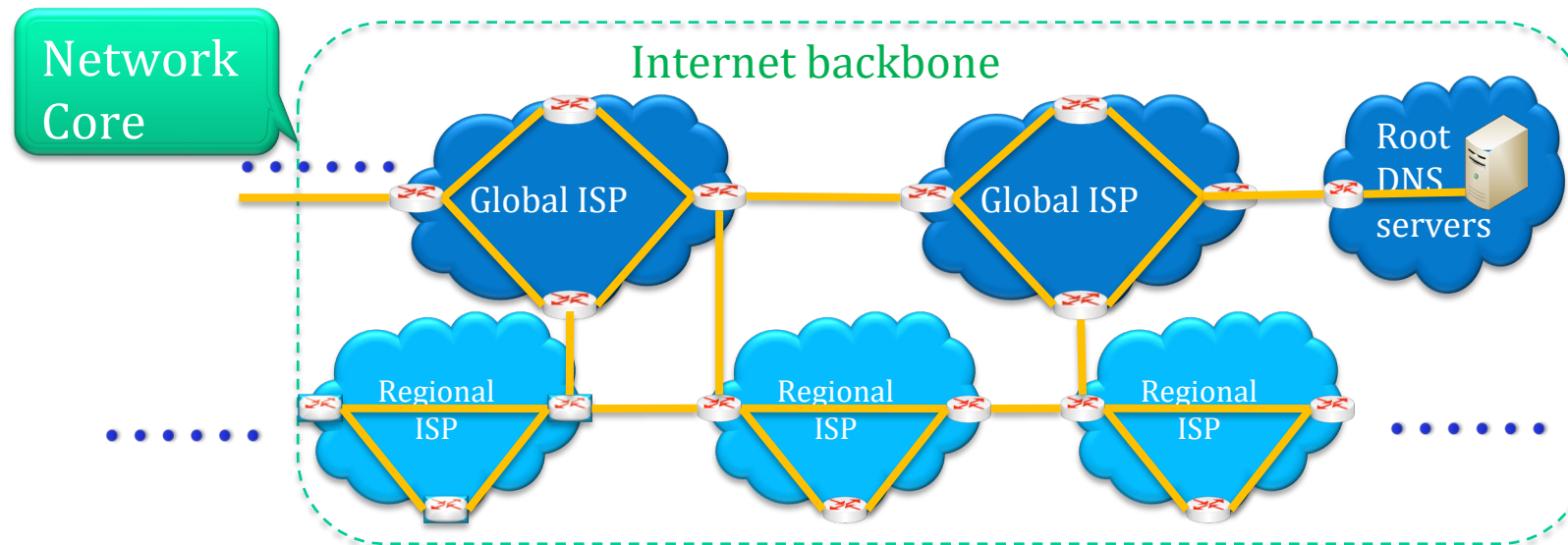
Computer Networks: The Network Core

- Routers and fiber links (in orange) form the Internet core
- Routers work together to figure out the most efficient path for routing a packet from source to destination host
 - A distributed algorithm can adapt to changing Internet conditions
 - Routing tables are generated and maintained in real-time



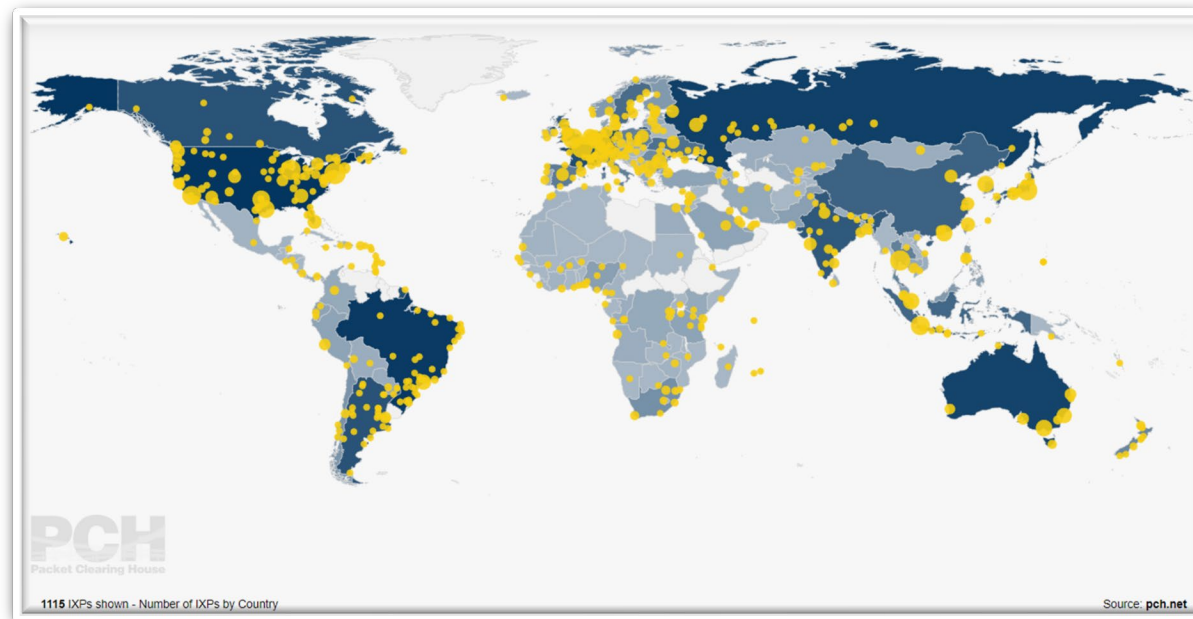
Computer Networks: The Network Core

- The core is provided by Internet Service Providers (ISPs) that interconnect multiple continents
- ISPs
 - Global ISPs or Tier-1 ISPs
 - Regional ISPs or Tier-2 ISPs



Computer Networks: Interconnected Network of Networks

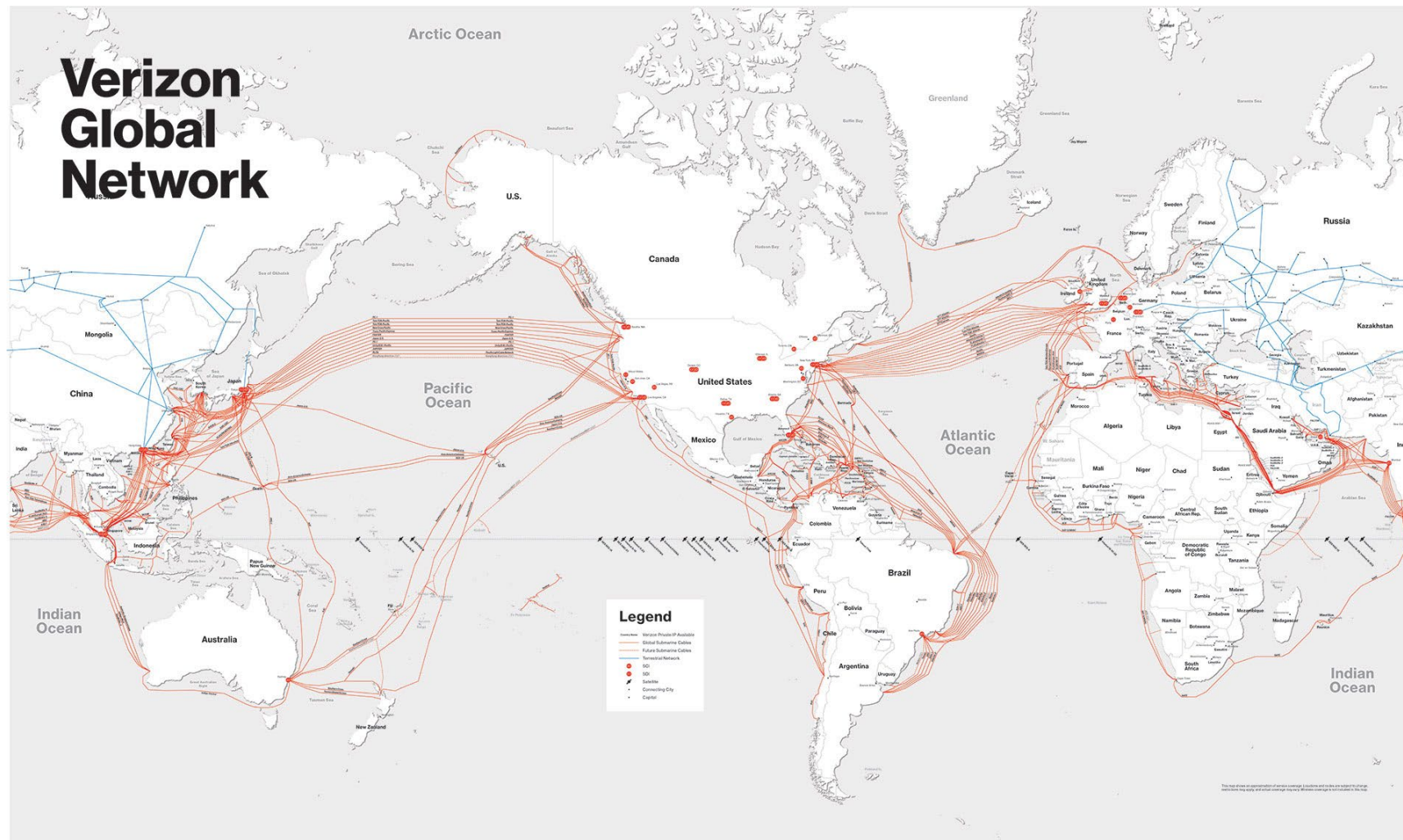
- Internet Backbone connects tier-1 ISPs
 - e.g., Verizon, Sprint, AT&T, Qwest, Level 3 Communications
 - The backbones of tier-1 ISPs are interconnected at various access points called Internet eXchange Points (IXP)
 - There were approximately 1,115 IXPs around the world (November 2021)
 - ✓ 218 in the U.S (the most IXPs in a country) and 15 in the Netherlands



Source: <https://www.pch.net/ixp/dir>

[Wu, Chwan-Hwa, & Irwin, J. David. (2013). Introduction to computer networks and cybersecurity. CRC Press.]

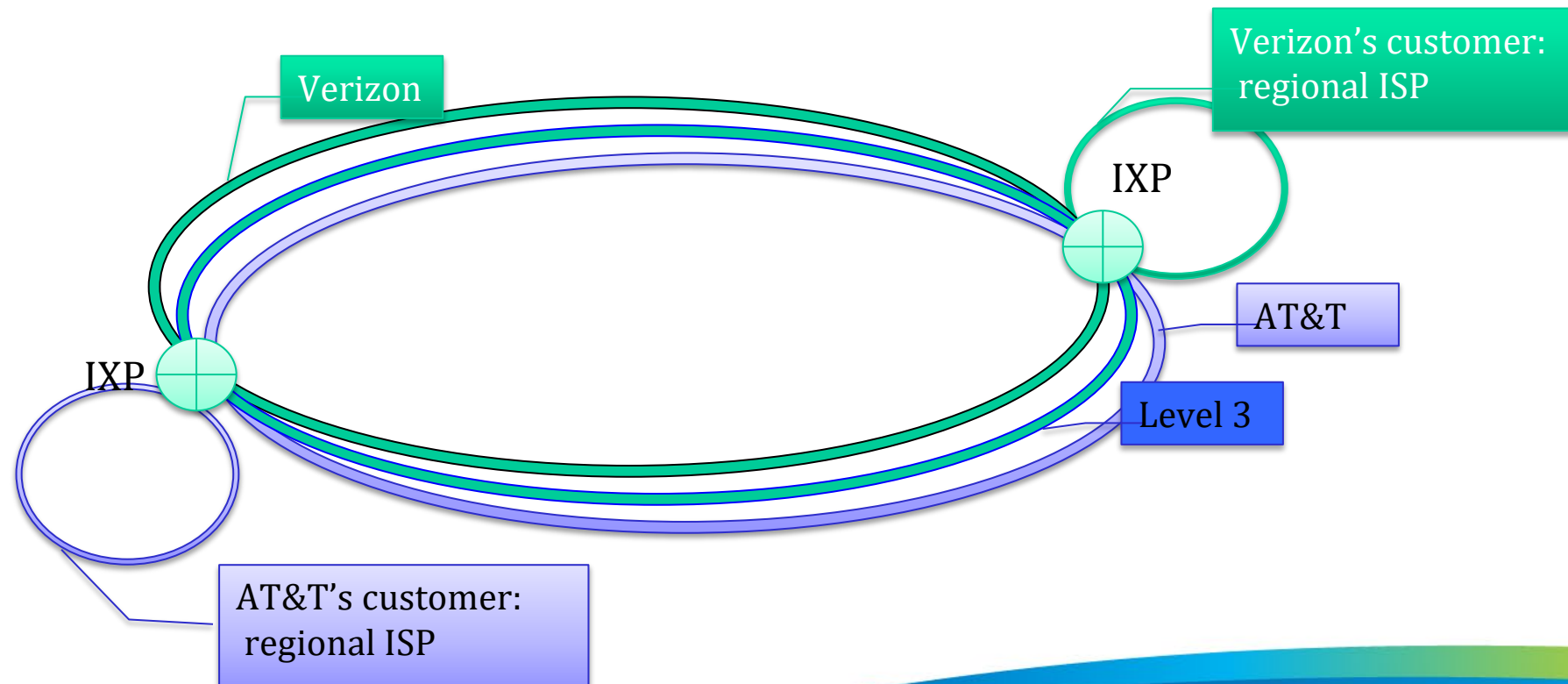
Computer Networks: Interconnected Network of Networks - Verizon Backbone



Source: <https://www.verizon.com/business/en-nl/why-verizon/>

Computer Networks: Hierarchical ISPs

- Regional ISPs: Tier-2 ISPs
 - Customer of Tier-1 ISP
 - Connect to one or more tier-1 ISPs, possibly to other tier-2 ISPs



Computer Networks: Packet Switching and Circuit Switching

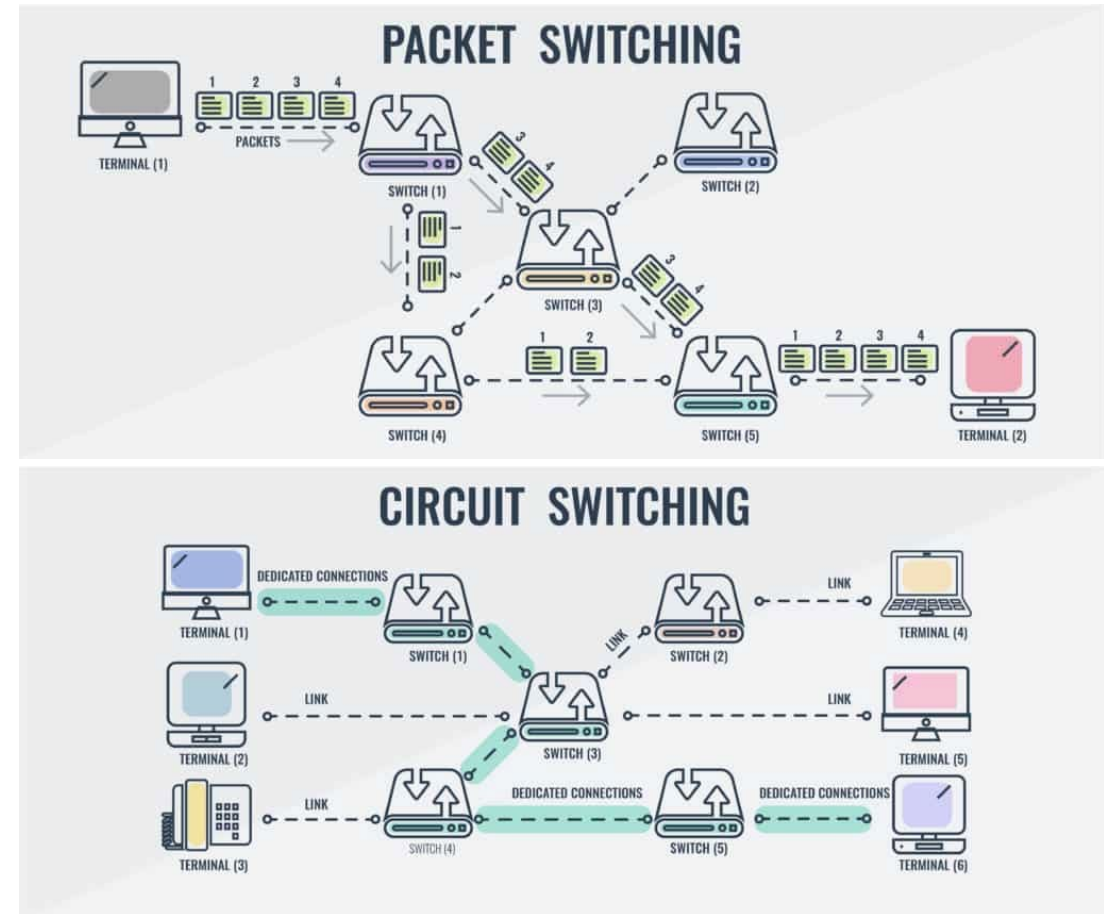
Data transfer on The Internet

- **Packet switching**

- Header contains source and destination IP address
- Best-effort delivery
 - ✓ Packets may be lost
 - ✓ Packets may be corrupted
 - ✓ Packets may be delivered out of order

- **Circuit switching**

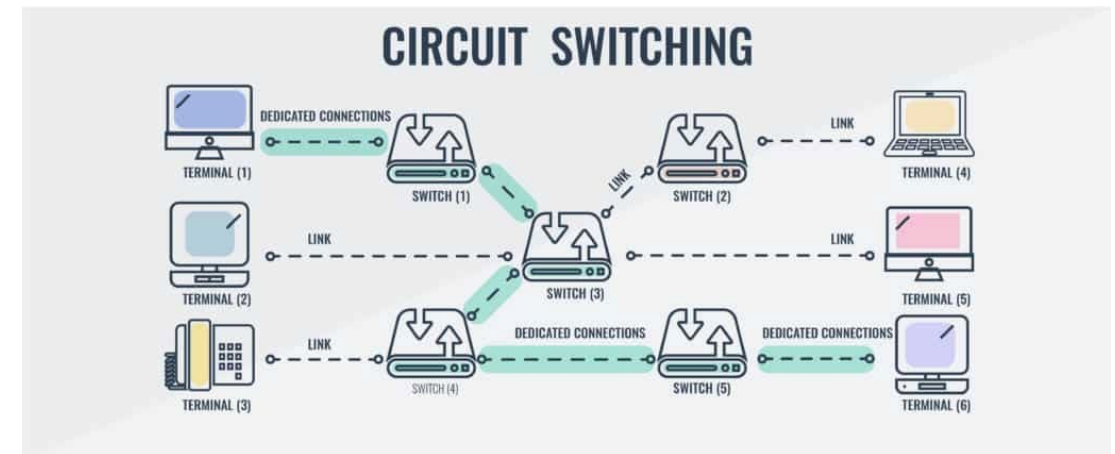
- Dedicated circuit per call: dial-up modem
- Virtual circuit:
 - ✓ Classical IP over ATM
 - ✓ Leased lines



Source: <https://www.comparitech.com/net-admin/circuit-switching-vs-packet-switching/>

Computer Networks: Circuit Switching

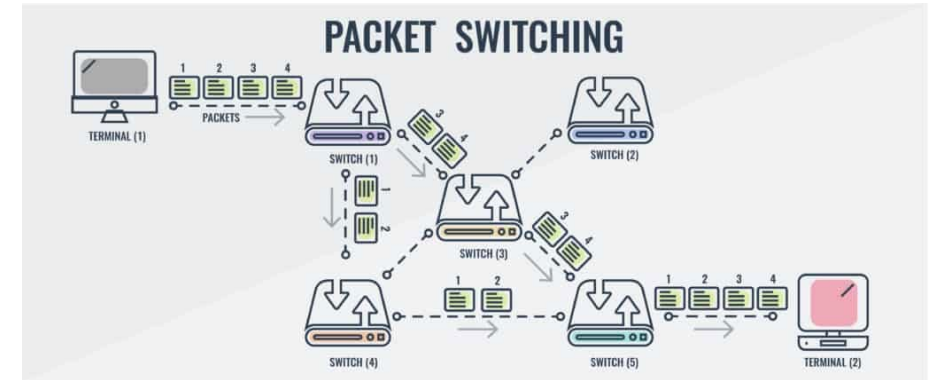
- End-to-end resources reserved for a connection
 - Establish a connection before data transfer
 - Circuit-like (guaranteed) performance
- Dedicated link bandwidth, and switch circuit capacity
 - Dedicated resources: no sharing
 - ✓ Resource is idle if not used by the call's owner
 - ✓ No queueing delay
 - Call setup and teardown required



Source: <https://www.comparitech.com/net-admin/circuit-switching-vs-packet-switching/>

Computer Networks: Packet Switching

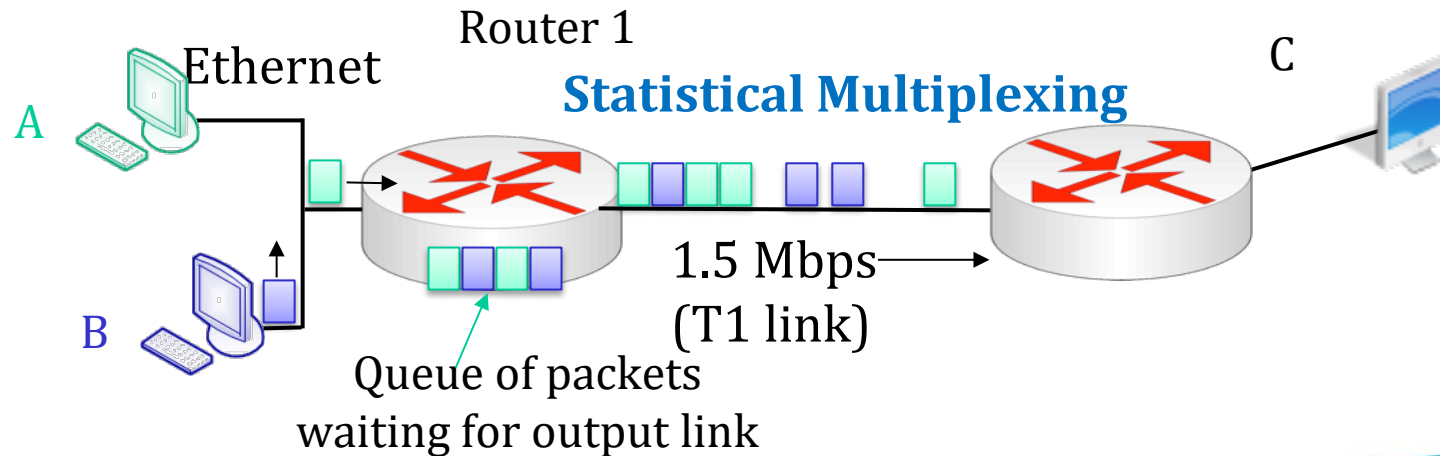
- The base of the Internet
 - Invented by Leonard Kleinrock, 1961
- Each host data stream is segmented into packets
- Destination IP address is in the packet header
 - Analogy: envelope address
- Each packet travels by itself using available resources provided by routers
- Packets may be lost while traveling
- Received packets may be out of order
- Transport layer must put the received packets in the correct order and reassemble them at destination



Source: <https://www.comparitech.com/net-admin/circuit-switching-vs-packet-switching/>

Computer Networks: Statistical Multiplexing

- Packet switching is based on statistical multiplexing
 - Packets are sent by hosts A and B
 - If there is no priority set, then every packet is treated the same based on the order of arrival
 - Router 1's T1 link bandwidth is shared by packets from hosts A and B
- In contrast, Frequency Division Multiplexing (FDM) and Time Division Multiplexing (TDM) let each host use the same slot and there is no resource contention



Computer Networks: Packet Switching vs Circuit Switching

- Packet switching is great for bursty data
 - Best effort delivery
 - Better for resource sharing
 - Network congestion
 - ✓ Packet delay in the router's queue
 - ✓ Packet loss due to queue overflow
 - Protocols need to pay overhead for reliable data transfer, congestion and flow control
- Circuit switching is great for voice and video
 - Bandwidth guarantee
 - Timing, latency, and latency jitter guarantee
 - Complex mechanisms lead to high cost for equipment and management

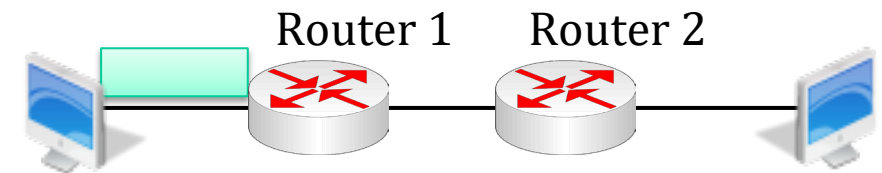
Computer Networks: Packet Switching Delays and Congestion

- Latency in packet switching network

- Store and forward packets in a router
- If the link is available, it takes L/R seconds to transmit one packet from Router 1 to Router 2
 - ✓ Transmission delay
- Assuming store and forward routing, entire packet must arrive at one router before it can be transmitted on the next link
- Host to host delay = $3L/R$ + propagation delay + other delays

- Example:

- ✓ $L = 1000$ Mbits
- ✓ $R = 100$ Mbps (Ethernet)
- ✓ Transmission delay per link = 10 sec
- ✓ **Total transmission delay = 30 sec**



Packet length: L bits
Link Rate: R bps

Computer Networks: Packet Switching Delays and Congestion

- Congestion and flow control

- Source host sends out too much information
 - ✓ Destination host cannot digest
 - ✓ Busy link
 - ✓ Router/switch buffer overflow
- Loss and delay causes packets to be resent in TCP
- Negative feedback causes even more congestion
- Congestion control throttles the output rate of source hosts to relieve congestion by using the symptom of congestion
 - ✓ Packet loss
 - ✓ Packet delay
 - ✓ Buffer overflow
- Flow control tells the source host how much information that destination host can digest
- Goal: to optimise the throughput rate (bits/sec) between source and destination

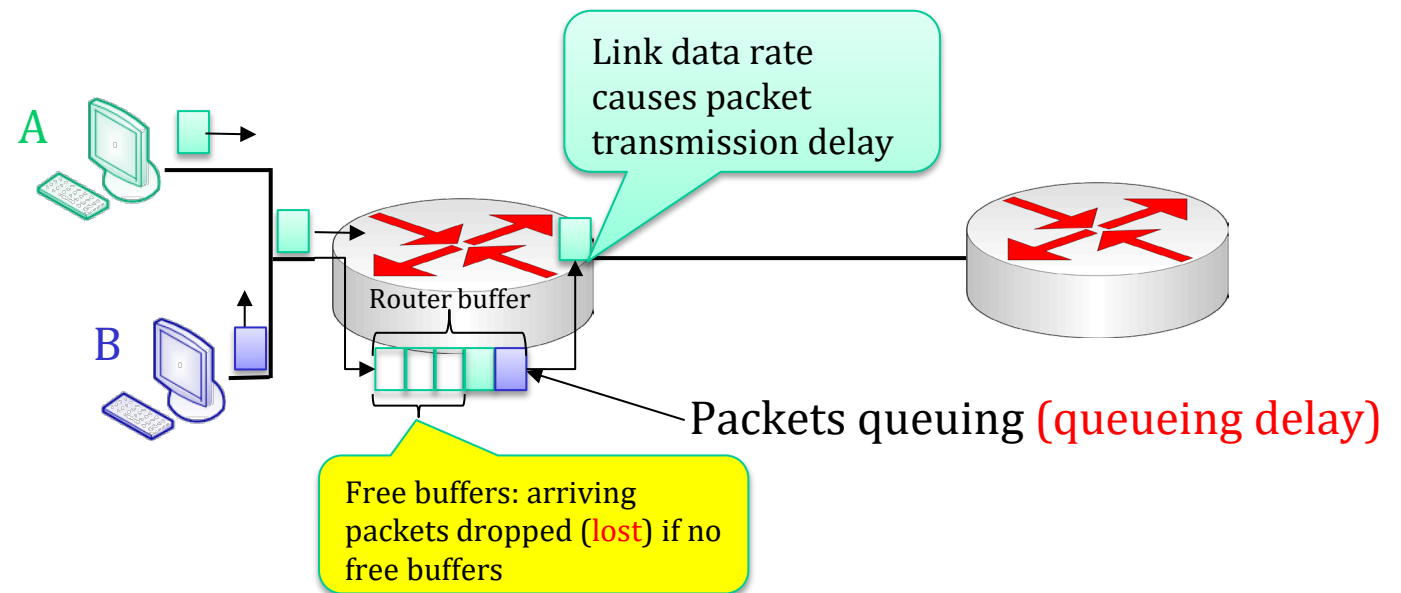


Source: frotcom.com

Computer Networks: Packet Switching Delays and Congestion

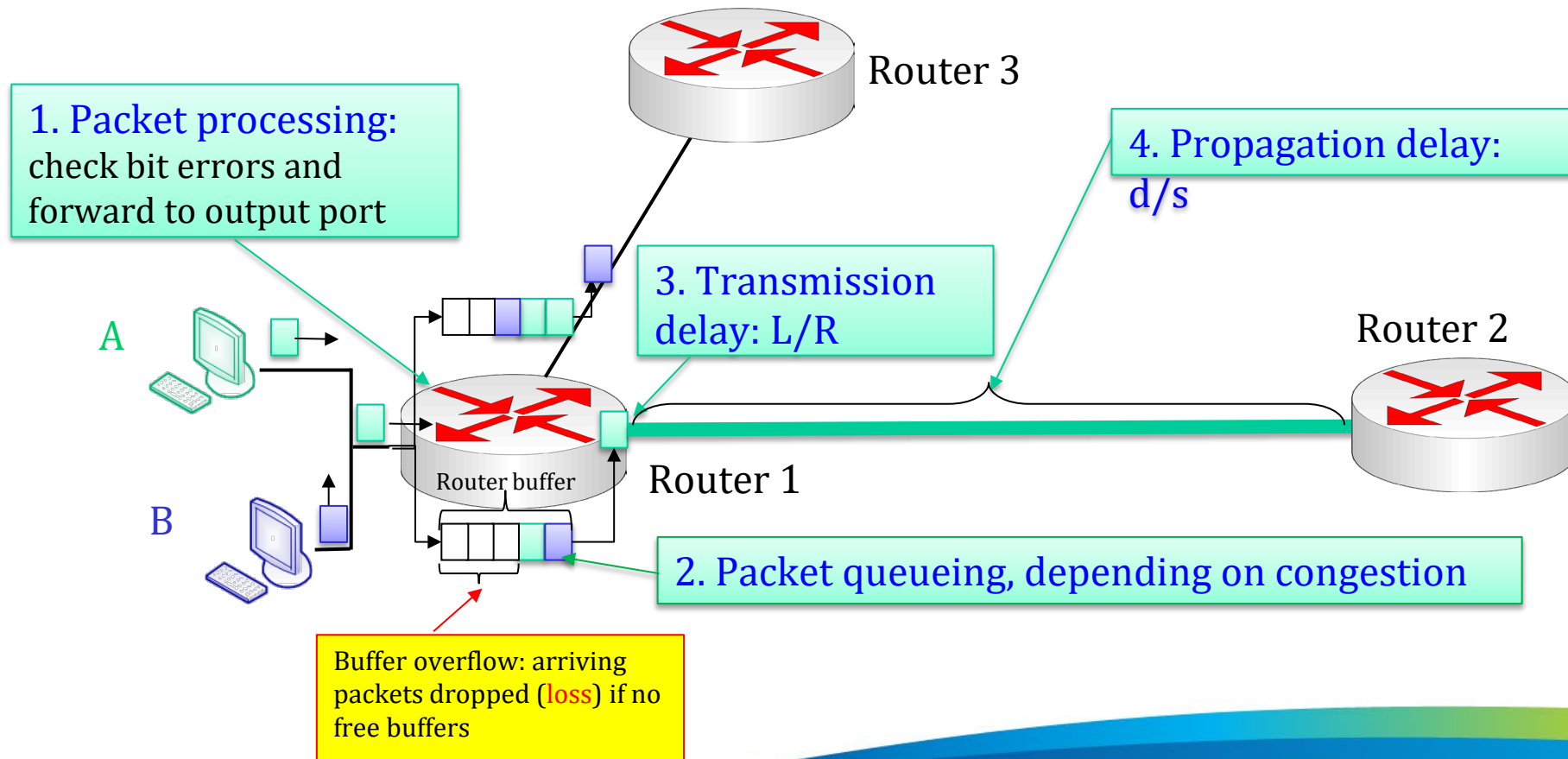
Packet loss and delay

- Packets queue in router buffers
 - Waiting for available output link with a finite pipe
 - Packet arrival rate exceeds output link capacity
 - ✓ Packets queue, and wait their turn
 - If the downstream links are busy (congested)
 - ✓ Queueing delay will be dominant
- If there is no free queue (buffer) space
 - Arriving packet is dropped (packet loss)



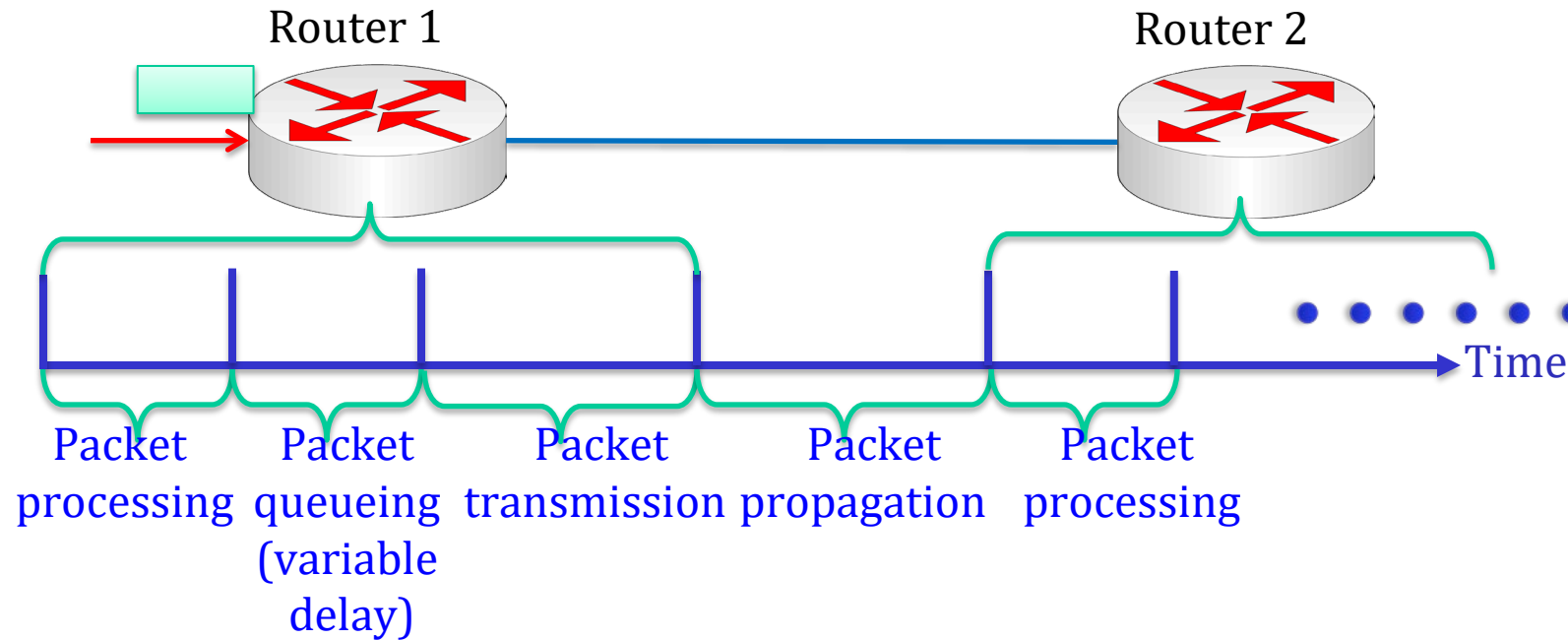
Computer Networks: Packet Switching Delays and Congestion

Total Delay of a node = processing delay + transmission delay + propagation delay + queueing delay



Computer Networks: Packet Switching vs Circuit Switching

- Router delays



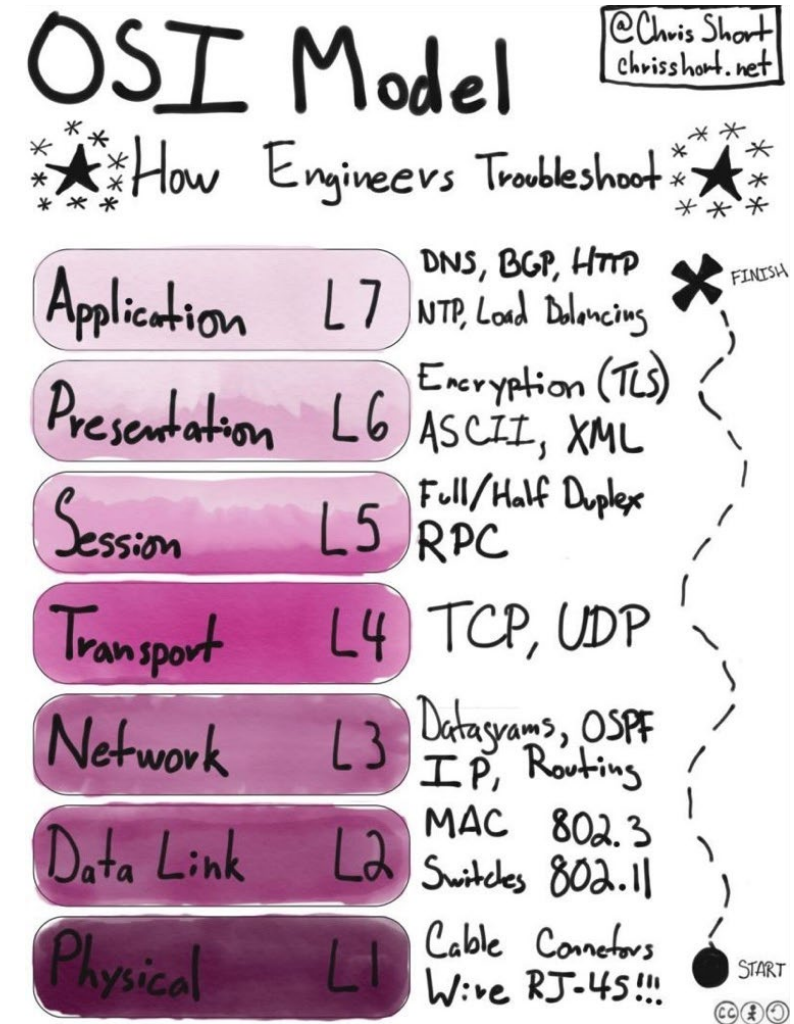
Computer Networks: Packet Switching vs Circuit Switching

- Packet switching cannot maintain the order of the packets
 - When packets travel on The Internet
 - ✓ They must pass through many routers
 - Today's Internet backbone routers typically employ multi-threaded network processors or Application-Specific Integrated Circuits (ASICs) for forwarding processing
 - Each router processes multiple packets in parallel
 - ✓ It is impossible to make sure that the output packets have the same order as the input packets due to parallel processing
 - Hence packet switching cannot maintain the order of the packets when a message contains multiple packets

Computer Networks: The Protocol Stack

- Modularization by layering

- Develop Internet: Divide and conquer by many people
- Internet can be modularized by layers
 - ✓ Open Systems Interconnection (OSI) model
- Modularization eases
 - ✓ Development (one company can tackle one module)
 - ✓ Maintenance
 - ✓ Updating the system
- Interaction between layers
 - ✓ Each layer relies on services from layer below
 - ✓ Each layer exports services to layer above
- Interface between layers defines interaction
 - ✓ Hides implementation details
 - ✓ Layers can change without disturbing other layers



Computer Networks: The Protocol Stack

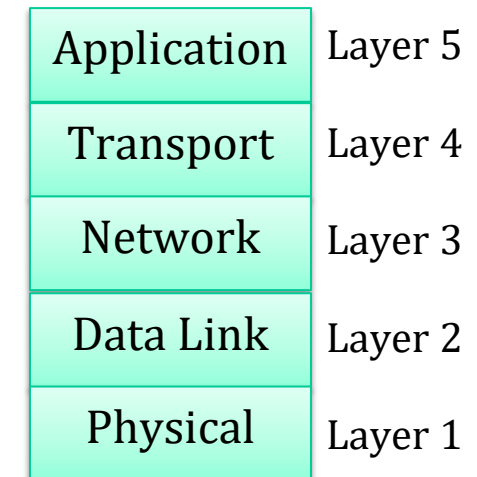
Internet protocol stack: US Department of Defense (DoD) Model

- Application Layer

- Network applications, web-based applications, socket programming using Java or C++
- HTTP, SMTP

- Transport Layer

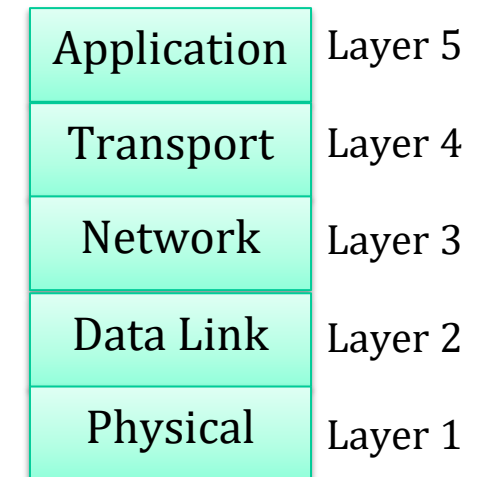
- Process-to-process data transfer
- Segment message at source and reassemble message at destination
- TCP: reliable transport with overhead
 - ✓ Transmission Control Protocol
- UDP: best effort delivery with little overhead
 - ✓ User Datagram Protocol
- SCTP: reliable transport based on transaction
 - ✓ Stream Control Transmission Protocol



Computer Networks: The Protocol Stack

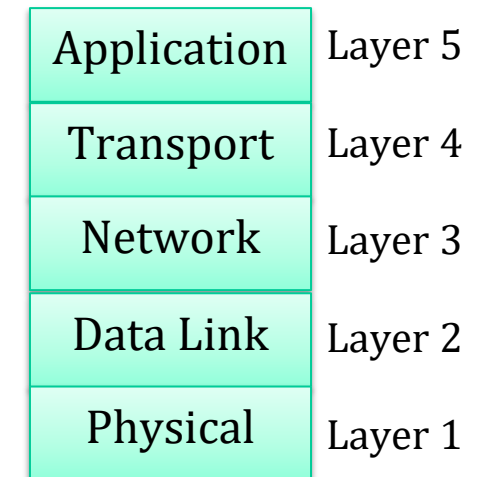
Internet protocol stack: US Department of Defense (DoD) Model

- Network Layer
 - Routing of datagrams from source subnet to destination subnet
 - IP, and routing protocols
- Data Link Layer
 - Data transfer between adjacent neighboring network nodes connected by a physical medium (copper, fiber or wireless)
 - Ethernet, WiFi
- Physical Layer
 - Bits propagate using copper, fiber and radio



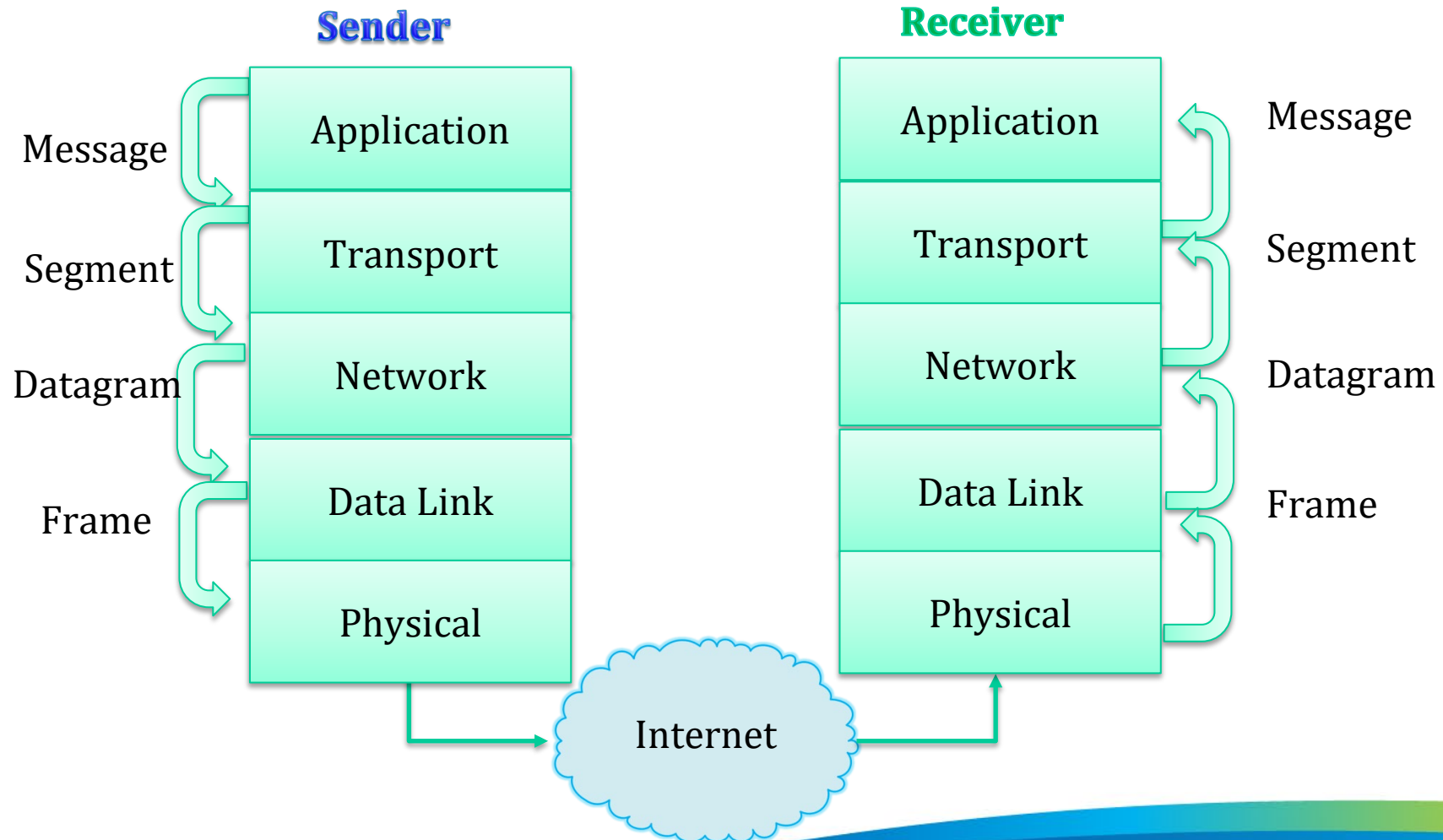
Computer Networks: The Protocol Stack

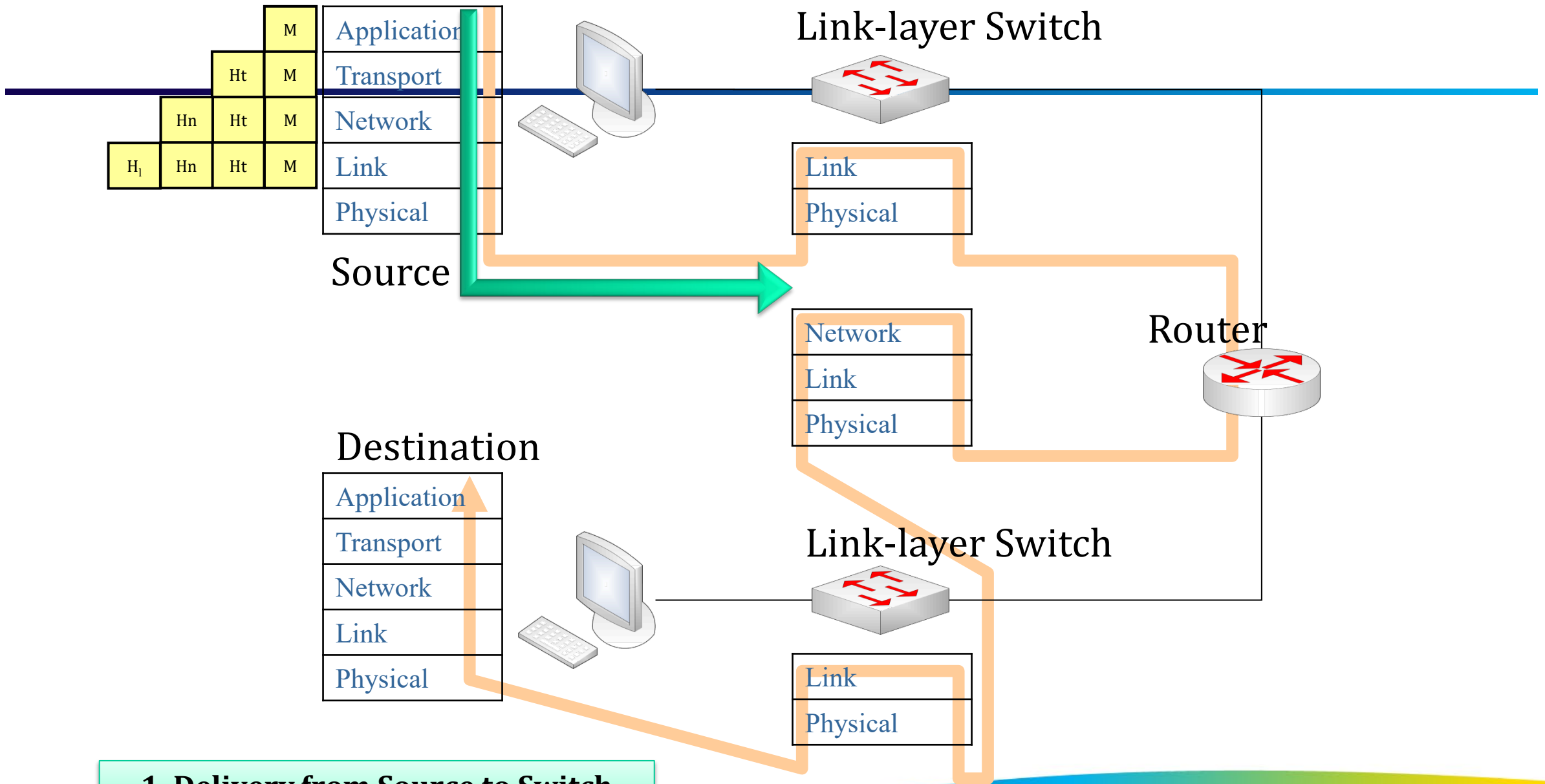
- Header for each layer
 - Each layer has a header
 - ✓ A header analogy is an envelope that contains source and destination addresses
 - Link layer has a header containing MAC (medium access control) addresses
 - Network layer has a header containing IP addresses
 - Transport layer has a header containing the port (service) number



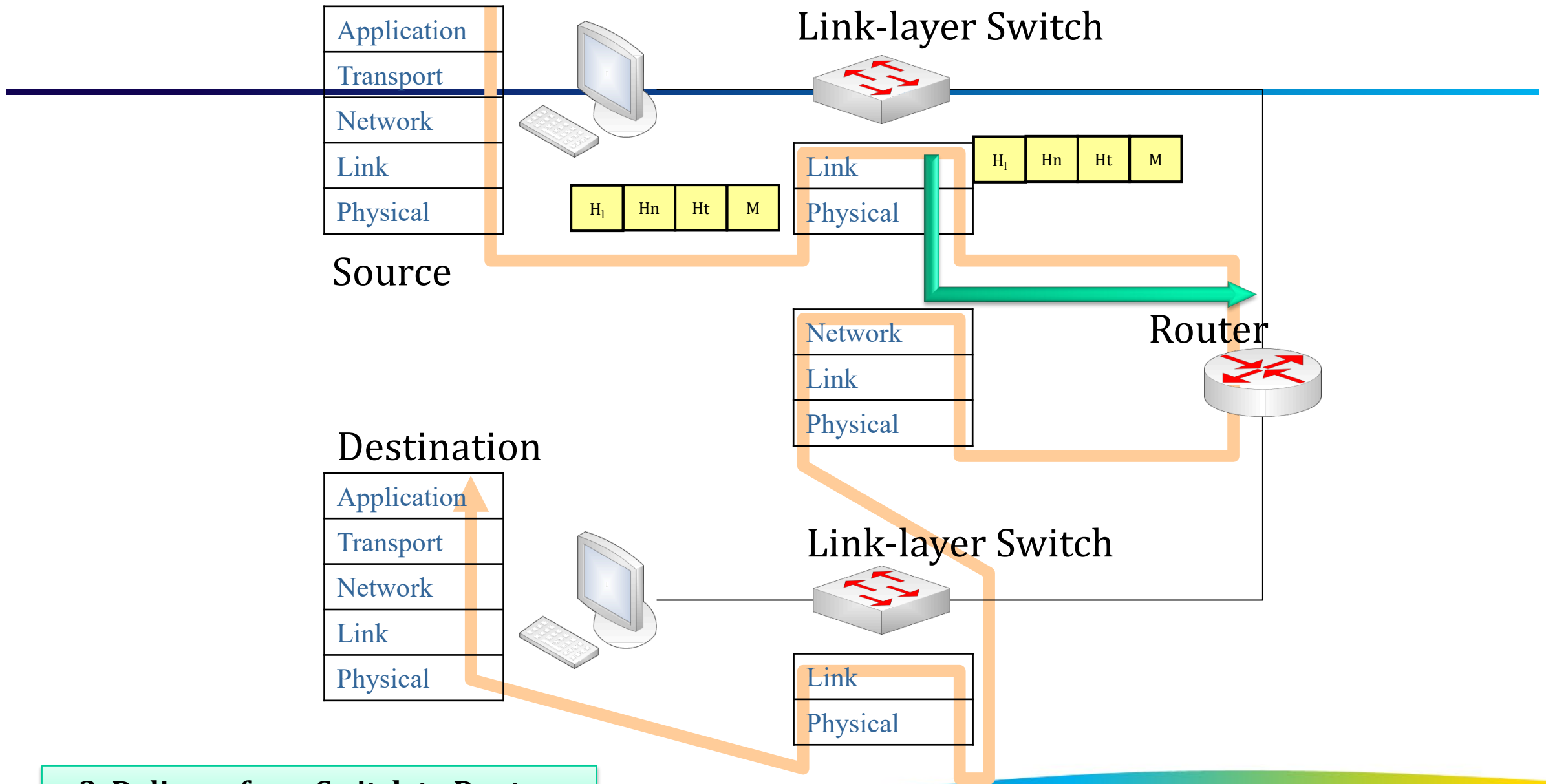
Computer Networks: The Protocol Stack

- Terms for bits processed in each layer

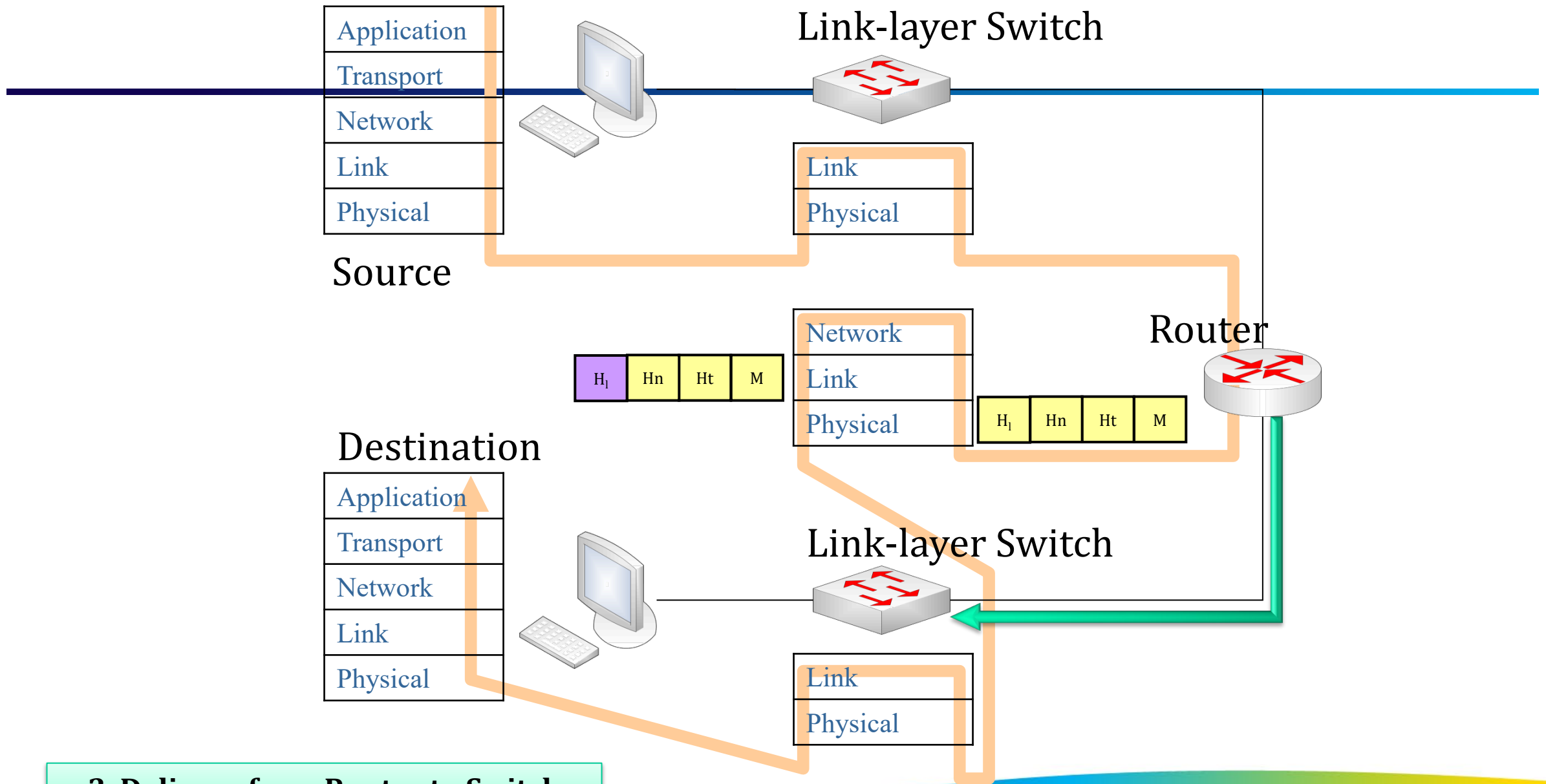




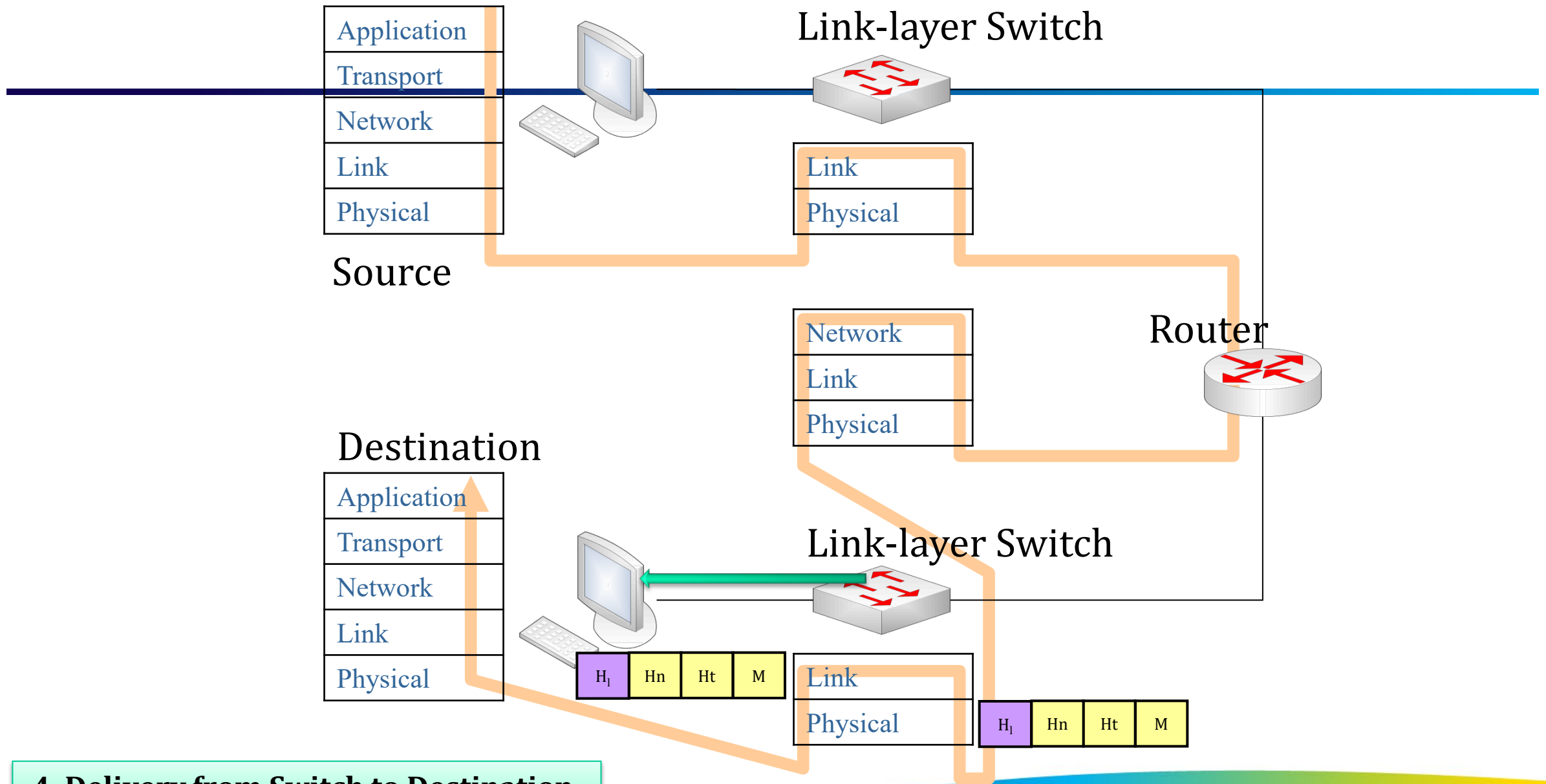
1. Delivery from Source to Switch



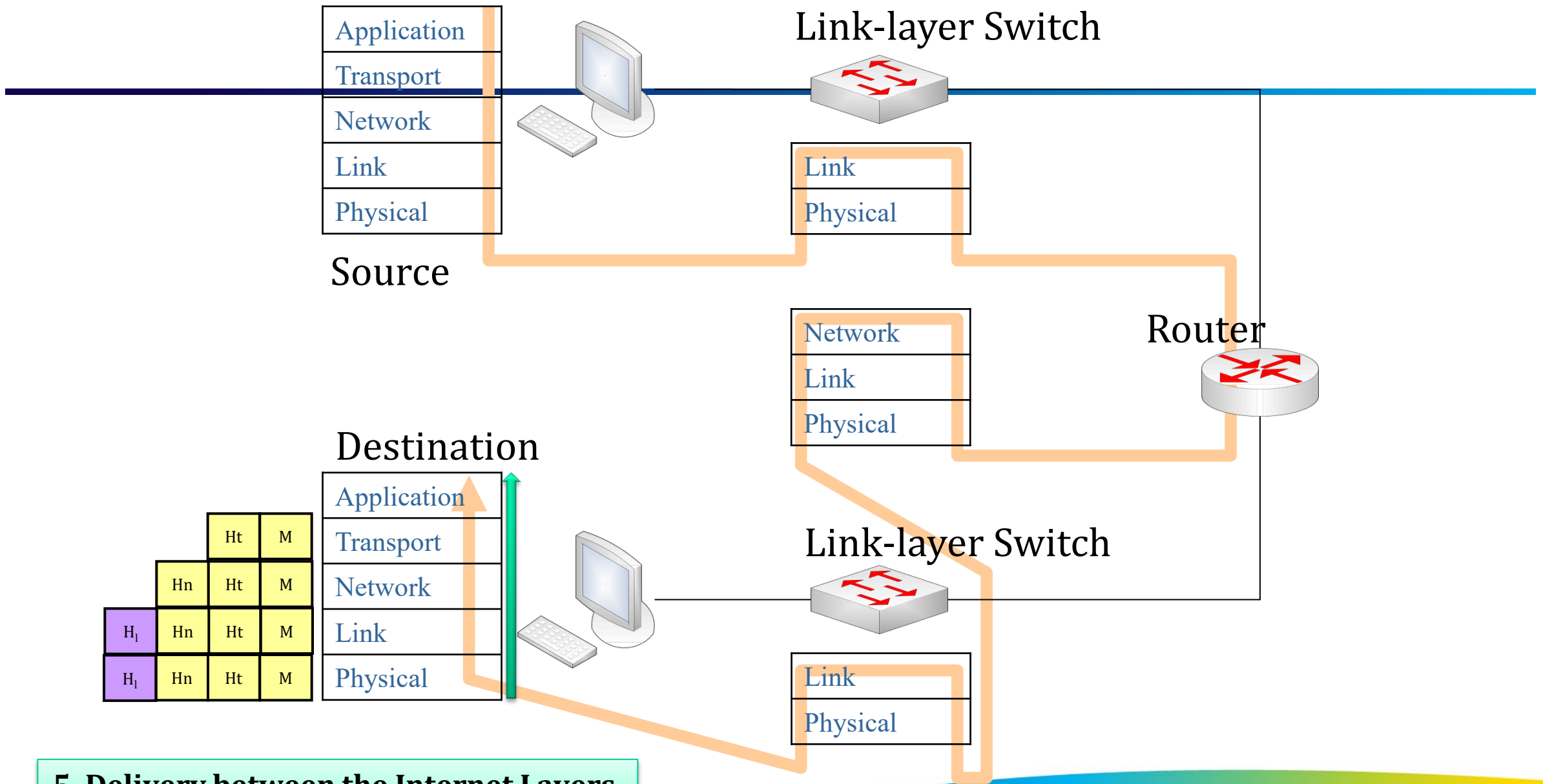
2. Delivery from Switch to Router.



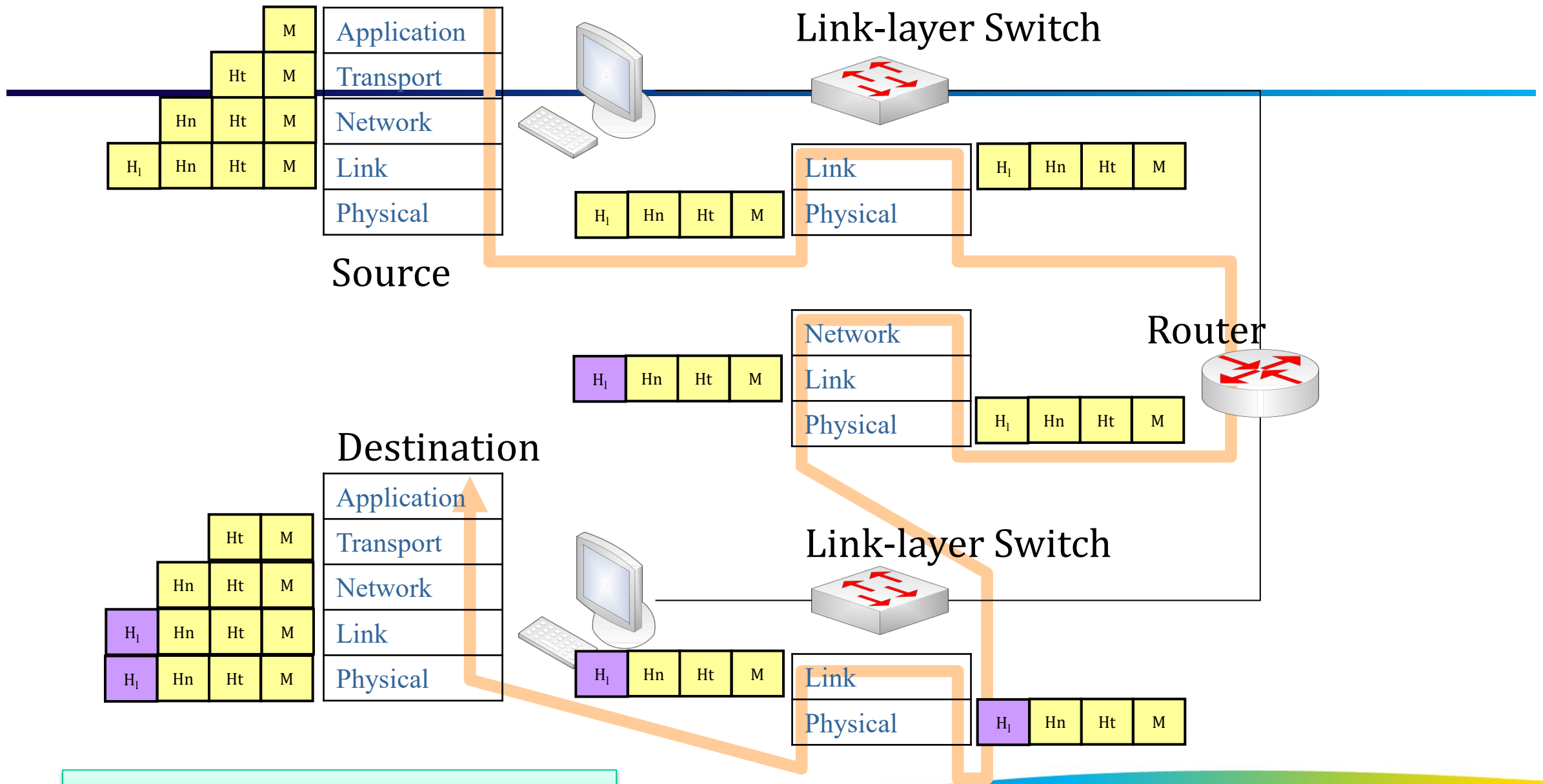
3. Delivery from Router to Switch



4. Delivery from Switch to Destination.



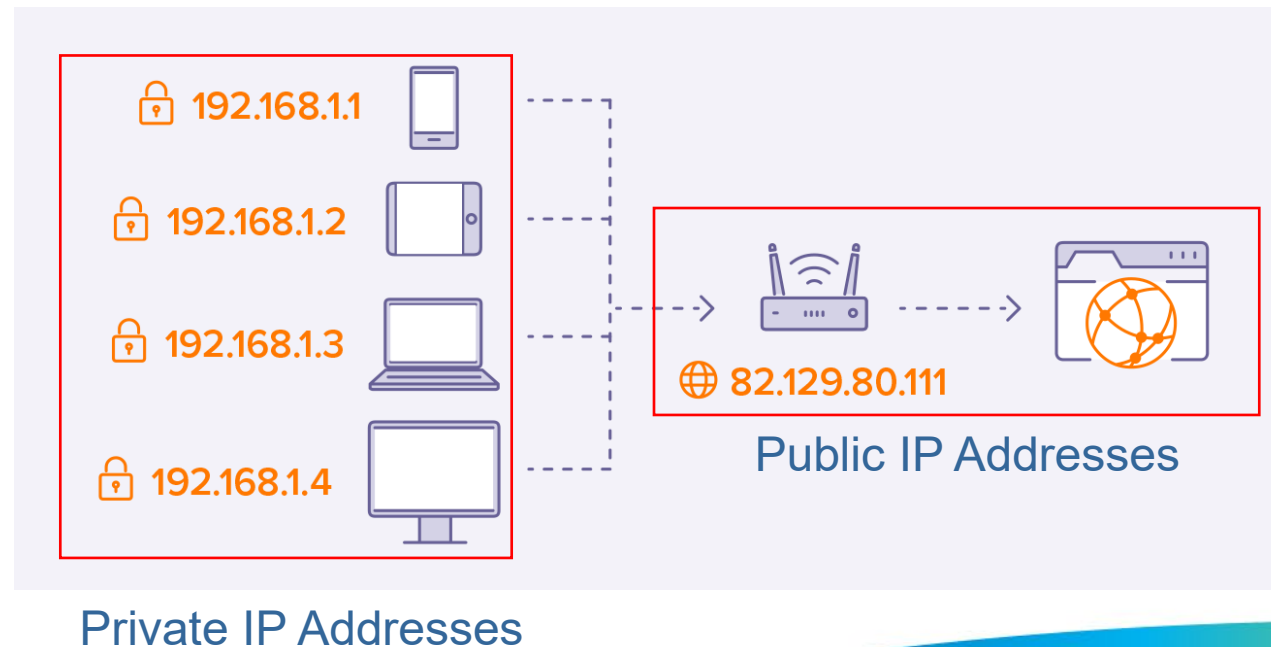
5. Delivery between the Internet Layers.



The entire delivery process.

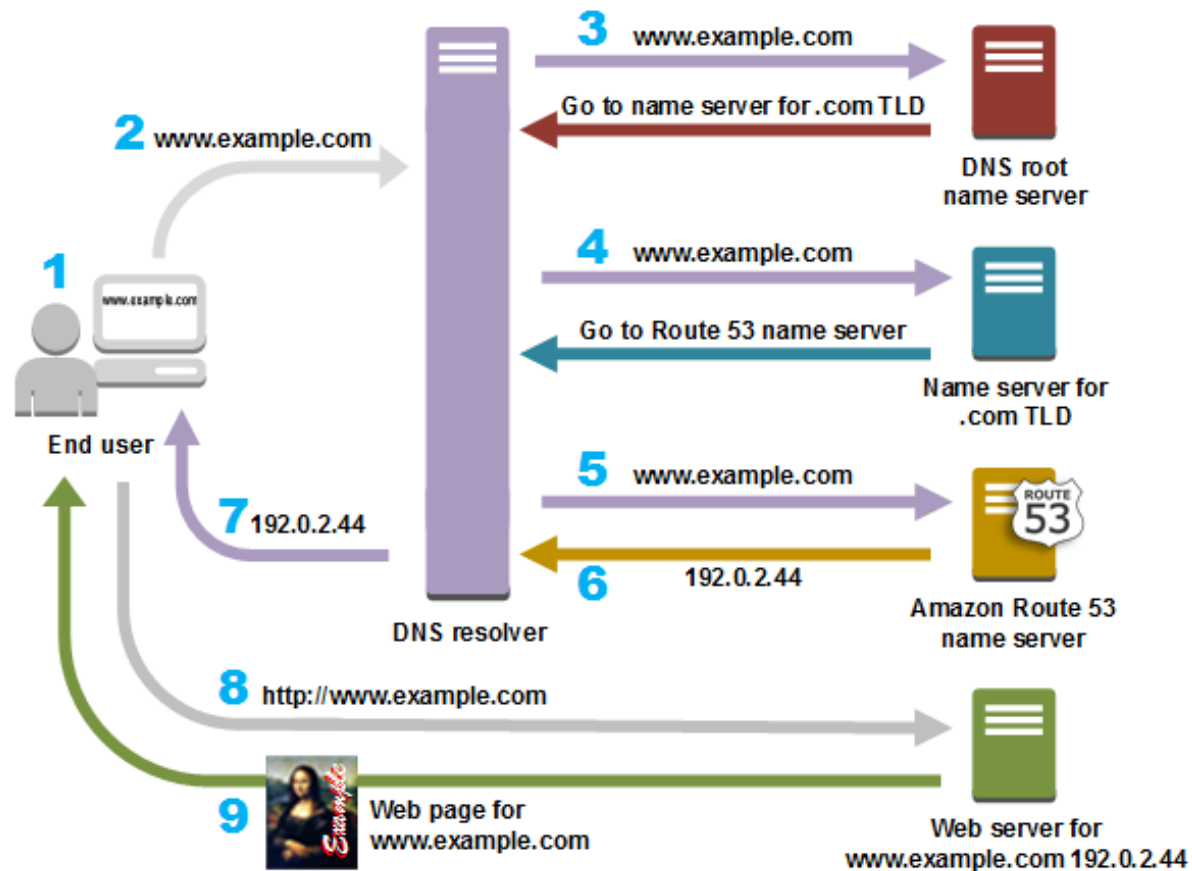
Public vs Private IP Addresses

- IP address is a unique address (number) that identifies a device on the Internet or a local network
 - A public IP address is an IP address that can be accessed directly over the Internet and is assigned to your network router by your Internet Service Provider (ISP)
 - A private (local) IP address is the address your network router assigns to your device



Domain Name System (DNS)

- DNS translates human readable domain names (example: `www.tudelft.nl`) to machine readable IP addresses (example: public IP `130.161.128.82`)



TU Delft website domain

```
Command Prompt
C:\Users>nslookup tudelft.nl
Server: UnKnown
Address: 192.168.0.1

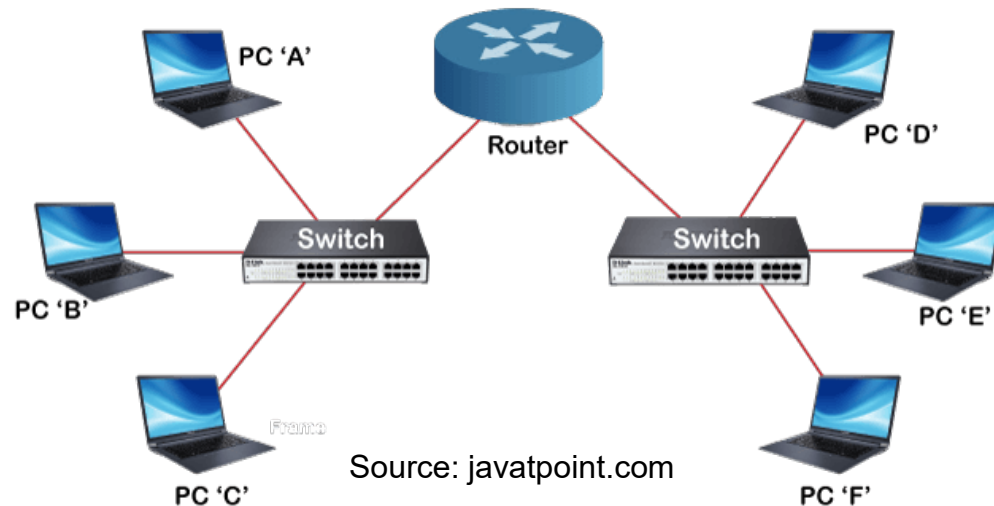
Name: tudelft.nl
Addresses: 2001:610:908:112:131:180:77:102
          130.161.128.82
```

IPv6 address

IPv4 address

Computer Networks: Router and Switch Functions

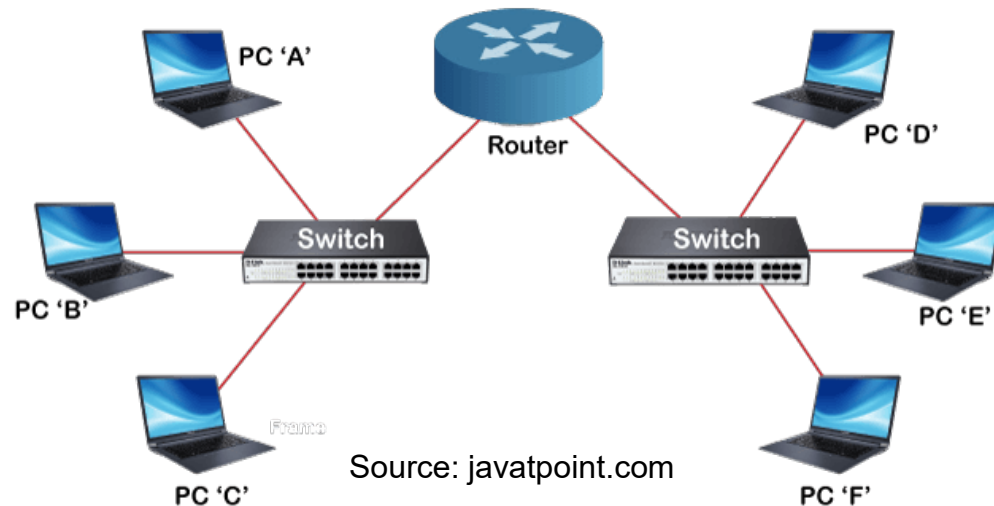
- A Layer 2 (Link-layer) switch cannot change the destination MAC address under any circumstance
- The Layer 2 switch, via a switching table, understands which port is designated by the destination MAC address. So, the switch can process the packet and direct it toward the correct port
 - ✓ The switching table is learned from each source's MAC address
- The source computer must know the IP address of the first router (gateway) and uses the Address Resolution Protocol (ARP) to obtain its MAC address
 - ✓ The destination MAC address of the packet leaving the source PC is the MAC address of the first router's interface.
 - ✓ However, the destination IP address is the destination PC's IP address



Connection of networks through Router

Computer Networks: Router and Switch Functions

- Routers and Layer 3 switches understand both IP addresses and MAC addresses
 - ✓ Routers work together to generate routing tables
 - ✓ A router or layer 3 switch knows the next hop's IP address from the routing table
 - ✓ The router then uses the ARP protocol to determine the MAC address of the destination PC (not layer 2 switch)
 - ✓ So, the destination MAC address, and the source MAC address, is changed by the router
 - ✓ Then the layer 2 switch, between the router and next PC, can switch correctly in accordance with the given MAC address
- The layer 2 switch learns the switching table from the source MAC address and the router learns the routing table from the IP address using the routing algorithm



Source: javatpoint.com

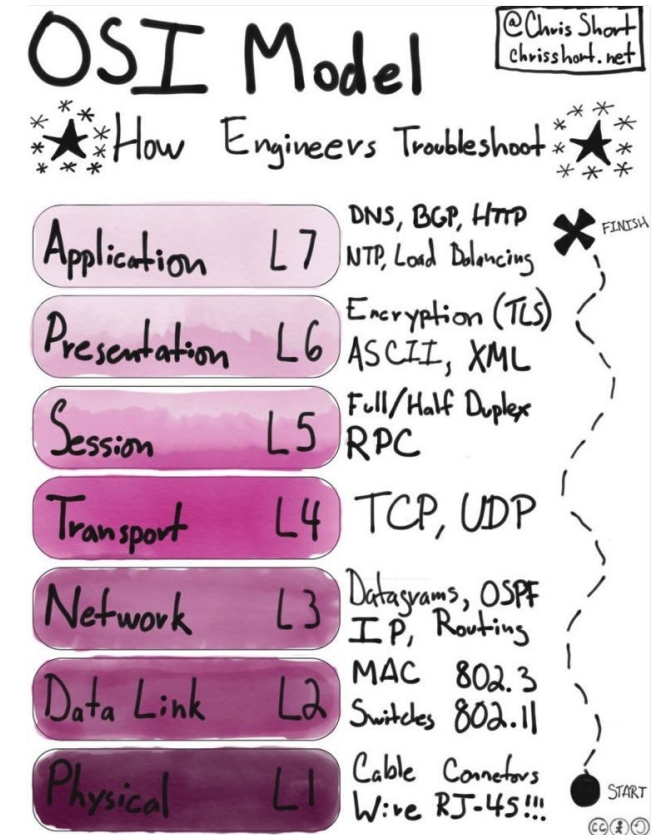
Connection of networks through Router

Computer Networks: Network Protocols

- All communication and activities on the Internet are governed by protocols
 - E.g., Dynamic Host Configuration Protocol (DHCP) provides a client with an IP address, gateway IP address and DNS IP address
- Protocols define
 - Packet format
 - ✓ Service (port) number in the TCP header, e.g., port 80 for http
 - ✓ Sequence and acknowledgement numbers are in the TCP header
 - The sequence of packets sent and received among network entities
 - Actions taken based on the parameters in the fields of a received packet
 - ✓ Retransmit of a packet depends on the acknowledgement number

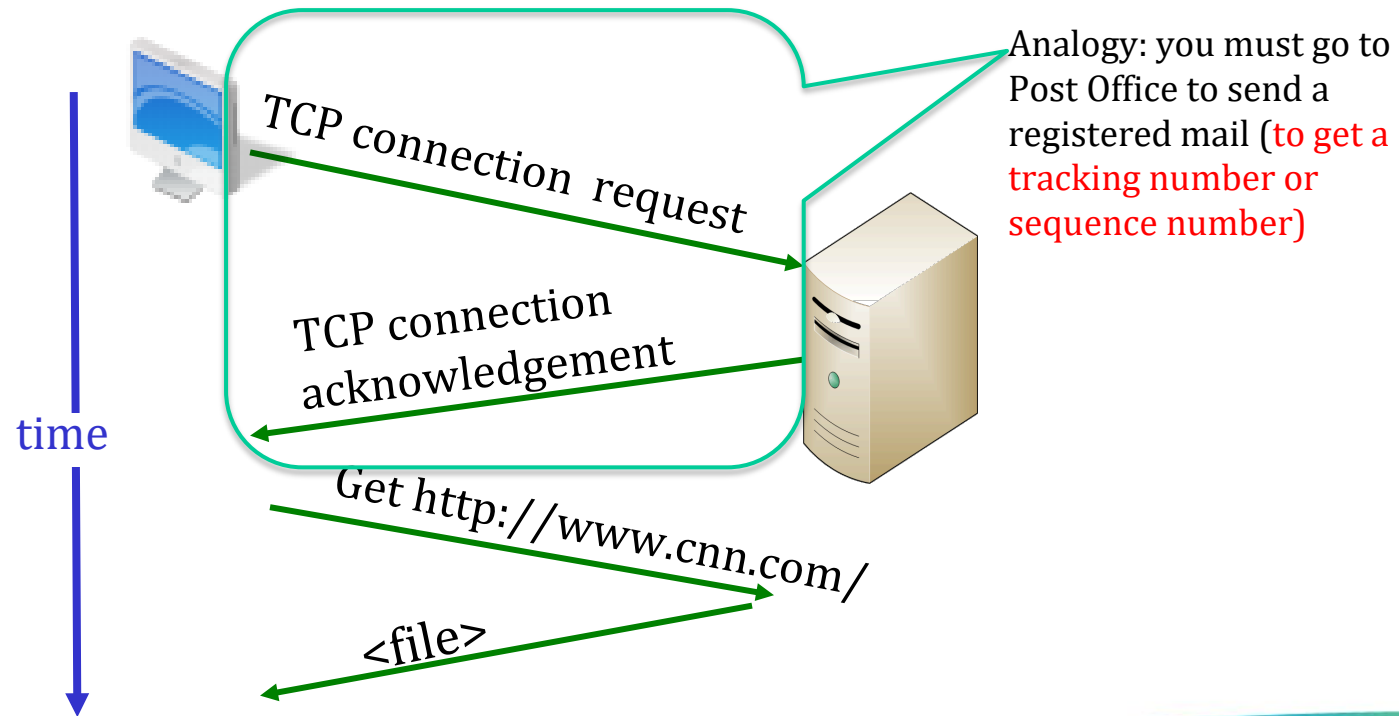
✿ Speaking the same language using the same standard set by IETF

✿ Syntax and semantics are critical



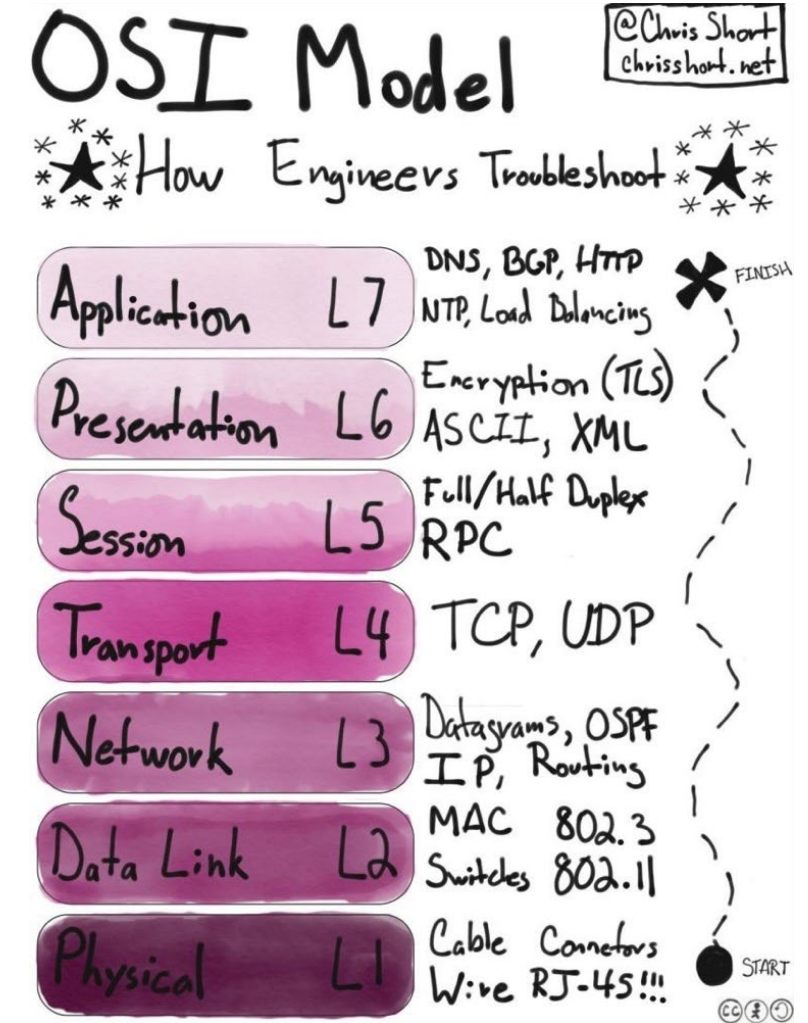
Computer Networks: Hypertext Transfer Protocol (HTTP) Protocol

- A connection is established between client and server
 - For reliable information delivery (aka connection-oriented service) using a packet sequence number
 - Connection can be established for the socket

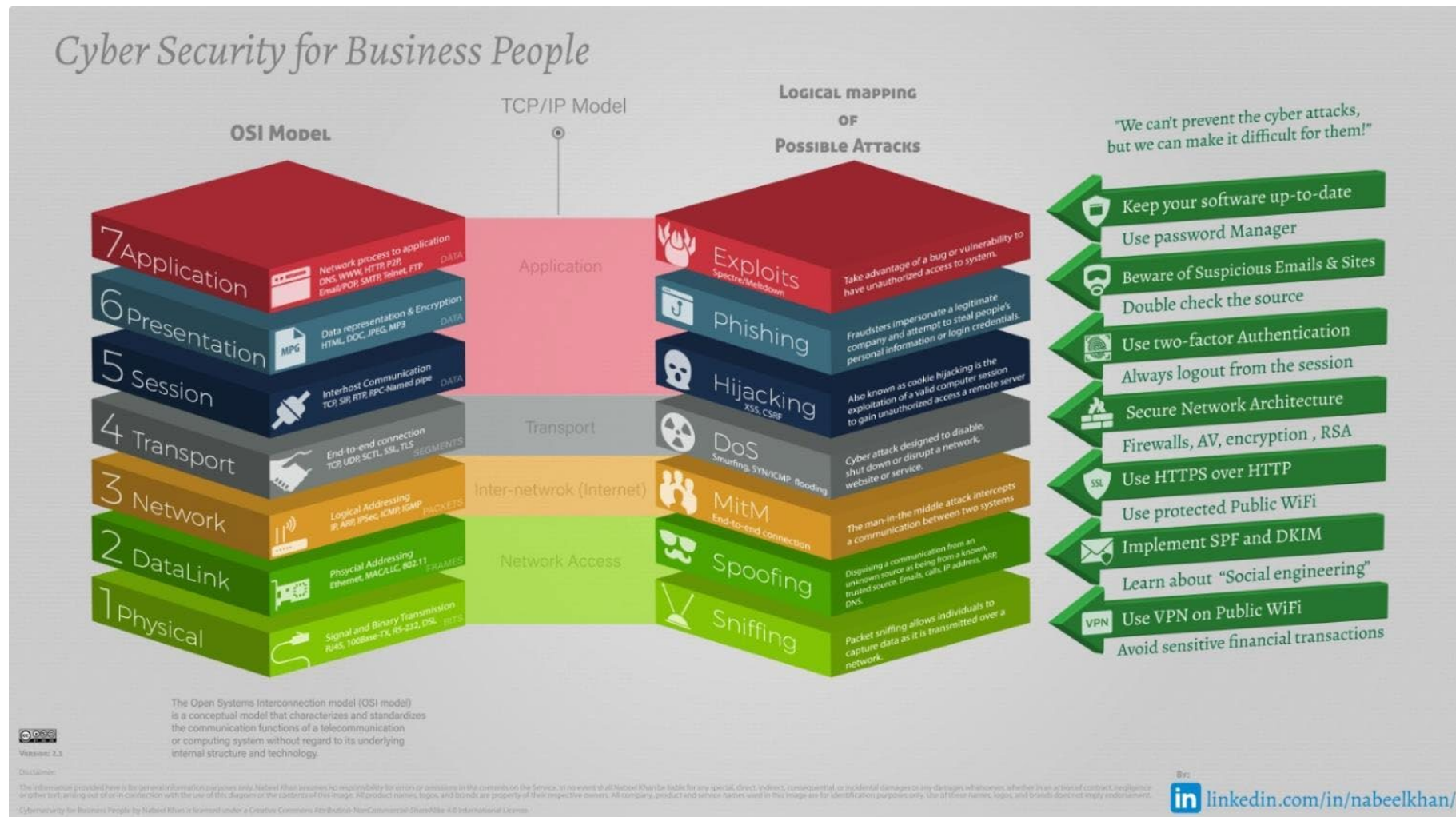


Computer Networks: Network Protocols

- Protocols (e.g., Ethernet 802.3, IP, TCP, HTTP) govern the following
 - Sending and receiving information packets according to standards
 - Taking actions specified in packets
 - Managing packet flow and congestion for optimal performance
 - Recovering lost packets
 - Forwarding packets from source host to destination host
- Example
 - Applications (such as HTTP) invoke transport protocols (TCP)
 - Transport protocols invoke the IP protocol
 - IP protocol invokes the Ethernet or equivalent protocol
 - Domain Name System (DNS) is used for naming
 - Glue between protocols (ARP, DHCP, ICMP)



Cyber Attacks on Protocol Stack and Cyber Security





GET IN
TOUCH

Alex Stefanov

Associate Professor, Chartered Engineer (CEng MIEI)

Email: A.I.Stefanov@tudelft.nl

Cyber Resilient Power Grids ([LinkedIn](#))

Control Room of the Future ([LinkedIn](#))

Intelligent Electrical Power Grids, EEMCS, **TU Delft**
Mekelweg 4, 2628 CD Delft, The Netherlands